

Technical paper

A hybrid multiobjective particle swarm optimization with Q-learning-driven local search for distributed heterogeneous hybrid flow-shop scheduling problem with worker–machine–environment collaboration

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ARTICLE INFO

Keywords:

Distributed hybrid flow-shop scheduling
Multiobjective particle swarm optimization
Q-learning-driven local search
Worker–machine–environment collaboration

ABSTRACT

The distributed heterogeneous hybrid flow-shop scheduling problem (DHHFSP) involves geographically dispersed factories, heterogeneous machines, and skilled workers, leading to complex multiobjective scheduling challenges. Existing studies usually ignore the critical role of workers in distributed manufacturing, and conventional multiobjective optimization algorithms struggle to balance convergence and solution diversity. To address these gaps, this paper develops a worker–machine–environment collaborative model for DHHFSP that simultaneously minimizes makespan, total energy consumption, and total worker cost, and proposes a hybrid multiobjective particle swarm optimization algorithm with Q-learning-driven local search (HMOPSO-QLS). The proposed approach features a hybrid framework that combines multidirectional particle swarm global search with reinforcement learning-based local search, a Pareto front oriented four directional swarm decomposition and update mechanism where three boundary exploration sub-swarms and one central exploration sub-swarm cooperatively guide particle evolution, and a two level local search scheme that integrates domain knowledge based inter-factory load balancing with Q-learning based adaptive intra-factory variable neighborhood search. Comprehensive experiments demonstrate that HMOPSO-QLS significantly outperforms classical multiobjective optimization algorithms in solving DHHFSP in terms of convergence and solution diversity. In distributed human–machine collaborative manufacturing, the proposed framework and algorithm support more effective configuration of factories, machines, and workers, providing robust schedules that are directly applicable to practical production decision-making and cross-factory coordination.

1. Introduction

With the increasingly fierce market competition and the rising demand for personalized products, traditional manufacturing paradigms have given way to more flexible and efficient production methods. The hybrid flow-shop scheduling problem (HFSP) model, which integrates parallel machines within a flow-line structure, has been widely adopted for its adaptability to multi-variety, small-batch production [1]. In the era of globalization, this has evolved into the distributed manufacturing system, where a network of cross-regional factories collaborates to enhance production flexibility and manage mass customization [2]. This shift has given rise to the distributed hybrid flow-shop scheduling

problem (DHFSP), a complex challenge involving the allocation of jobs to different factories and the subsequent optimization of job sequences and machine assignments within each facility.

Recent developments in manufacturing emphasize the transition from purely machine-centric automation to human-centric collaborative production, where workers and machines work synergistically to achieve sustainable and personalized production goals [3]. A smooth and harmonious human–robot collaboration is expected to achieve sustainable productivity and also expands workers' prosperity, an important indicator of social sustainability within production systems [4]. Simultaneously, the world is facing increasingly severe environmental

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<https://doi.org/10.1016/j.jmsy.2025.12.024>

Received 21 July 2025; Received in revised form 16 December 2025; Accepted 26 December 2025

Available online 31 December 2025

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and resource challenges, making sustainable development in manufacturing a core issue [5]. This paradigm shift requires modern scheduling strategies to balance production efficiency, environmental impact, and economic costs in order to realize sustainable manufacturing [6]. Modern scheduling strategies, as a key technology for industrial intelligence, promote green and high-quality development in the industry [7].

This human-centric and sustainability-oriented approach fundamentally challenges traditional scheduling formulations, where workers are often treated as a uniform and inexhaustible resource, or their role is entirely neglected in favor of machine-centric models [3]. In contrast, workers represent a critical and heterogeneous resource with varying skill levels, productivity rates, and associated costs. The integration of worker considerations into scheduling decisions has therefore become increasingly important as manufacturing systems seek to optimize not only machine utilization but also worker allocation and costs [8,9], motivating the development of worker-integrated scheduling models that explicitly incorporate worker resource constraints and capabilities as integral components of the optimization process.

The specific problem addressed in this study is the distributed heterogeneous hybrid flow-shop scheduling problem with integrated worker-machine-environment considerations (DHHFSP-WME), which represents a comprehensive optimization challenge in modern manufacturing systems. This problem is characterized by significant heterogeneity, where distributed factories may differ in their production lines, machine capabilities, and worker skill sets, complicating system-wide scheduling [10]. The core challenge lies in coordinating multi-dimensional resource allocation to balance conflicting objectives such as production efficiency, economic costs, and environmental impact in the presence of heterogeneous workers, machines, and factories.

The complexity of DHHFSP-WME necessitates sophisticated optimization methods to address its multi-dimensional characteristics. However, existing algorithms in multi-constraint, multi-factory environments struggle to simultaneously balance global exploration and solution set convergence, and traditional multiobjective optimization algorithms often generate unevenly distributed solution sets on the Pareto front, limiting scheduling flexibility when managers need alternative solutions in different trade-off regions. Meanwhile, distributed manufacturing involves dual optimization requirements: inter-factory load coordination and intra-factory human-machine collaboration. Existing local search strategies typically rely on general-purpose operators and static search mechanisms that are insufficiently problem-specific, making it difficult to adapt to factory heterogeneity and to capture the crucial role of worker skill matching during solution improvement.

To address these challenges, this study establishes a worker-machine-environment collaborative model with minimizing makespan, total energy consumption (TEC), and total worker cost (TWC) for DHHFSP-WME and proposes a hybrid multiobjective particle swarm optimization algorithm with Q-learning-driven local search (HMOPSO-QLS).

The main contributions of this work are as follows:

- (1) A hybrid algorithm framework that combines multi-directional global search strategies based on multi-directional particle swarm decomposition and update strategies with two-level reinforcement learning (RL)-driven local search mechanisms, specifically designed to handle distributed heterogeneous manufacturing complexity.
- (2) A Pareto front-oriented four-directional particle swarm decomposition approach that strategically divides the particle swarm into three boundary exploration sub-swarms and one central exploration sub-swarm with distinct update strategies, thereby significantly enhancing convergence and distribution performances.
- (3) A hierarchical two-level local search architecture that utilizes domain knowledge-driven operators for inter-factory load balancing while implementing Q-learning-based adaptive variable neighborhood search (VNS) for intra-factory optimization, significantly enhancing the comprehensive exploitation capability within the problem domain.

From a manufacturing systems perspective, the proposed model and HMOPSO-QLS framework coordinate global and local decisions across distributed factories, heterogeneous machines, and skilled workers, and use multidirectional multiobjective search to provide well distributed trade-offs among makespan, TEC, and TWC. The hierarchical two-level local search, combining domain knowledge driven inter-factory load balancing with Q-learning-based intra-factory VNS, aligns with actual cross-factory planning and shop floor rescheduling processes. As a result, the approach not only fills research gaps regarding workers and distributed collaboration, but also offers practically implementable schedules for improving production efficiency, economic cost, and labor utilization.

The remainder of this study is organized as follows: Section 2 provides a comprehensive review of related work. Section 3 presents the mathematical formulation of the DHHFSP-WME. Section 4 details the proposed HMOPSO-QLS algorithm and its key components. Section 5 presents comprehensive experimental results and comparative analysis. Section 6 discusses the limitations of the current study and outlines directions for future research. Finally, Section 7 summarizes the main contributions and findings of this work.

2. Related work

The DHFSP lies at the intersection of distributed manufacturing and complex shop-floor optimization. Existing research has evolved from production-centric models to green scheduling that accounts for energy efficiency, and more recently to approaches that explicitly incorporate human factors. This section critically reviews these three research streams to position our work and identify the specific gaps it addresses.

2.1. Production and green-oriented DHFSP

The inherent complexity of DHFSP, characterized by its distributed architecture and multi-stage hybrid flow, necessitates sophisticated optimization techniques. Most studies focus on bi-objective optimization, targeting classical production metrics and environmental considerations. Makespan is the dominant efficiency indicator in existing works, while tardiness and earliness receive comparatively less attention.

Early research concentrated on production-oriented objectives, with Lu et al. [11] and Zheng et al. [12] establishing foundational bi-objective approaches. Subsequent studies by Cai et al. [13] and Jiang et al. [14] incorporated additional real-world constraints such as sequence-dependent setup times and fuzzy processing times, thereby refining production-oriented models.

Driven by global pressure for sustainable manufacturing and stricter environmental regulations, green-oriented DHFSP research has grown rapidly. Recent studies widely incorporate environmental objectives, typically formulating bi-objective optimization of makespan and TEC to balance productivity and energy efficiency, as seen in [15–17].

Various optimization algorithms have been employed, each showing different strengths across DHFSP variants. Evolutionary algorithms dominate, with genetic algorithms and their variants demonstrating consistent effectiveness in diverse formulations [14–16,18,19]. Swarm intelligence methods, particularly particle swarm optimization (PSO) and ant colony optimization (ACO), constitute another major category with robust performance [13,20,21]. Iterated greedy algorithms exhibit strong performance on small- and medium-scale instances but face scalability challenges [11,17,22]. Cooperative evolution approaches, though less frequent, perform well in complex multi-factory coordination scenarios [12,23,24]. Alternative techniques, such as teaching-learning-based optimization (TLBO) [25], provide tailored solutions for specific variants. More recently, integration of RL has emerged as an important trend: studies in [16,20,21,24,26] show that RL-hybrid methods enhance solution quality via adaptive operator selection, search policy design, or cooperative evolution frameworks with deep RL components.

Table 1
Literature on multiobjective production and green-oriented DHFSP.

Ref.	Objectives						Algorithms						Factory	
	Mksp.	Tard.	Earli.	Env.	Worker	Others	EA	SI	Co-evol.	Greedy	with RL	Others	Homo.	Hetero.
[11]	✓	–	–	✓	–	–	–	–	–	✓	–	–	–	✓
[12]	✓	–	–	–	–	Robustness	–	–	✓	–	–	–	✓	–
[13]	✓	✓	–	–	–	–	–	✓	–	–	–	–	✓	–
[14]	✓	–	–	✓	–	–	✓	–	–	–	–	–	✓	–
[15]	✓	–	–	✓	–	–	✓	–	–	–	–	–	–	✓
[16]	✓	–	–	✓	–	–	✓	–	–	–	✓	–	–	✓
[17]	✓	–	–	✓	–	–	–	–	–	✓	✓	–	✓	–
[30]	–	✓	–	✓	–	–	✓	–	–	–	–	–	–	✓
[25]	✓	✓	–	–	–	–	–	–	–	–	–	TLBO	✓	–
[31]	✓	–	–	–	–	transp. cost	✓	–	–	–	–	–	–	✓
[20]	✓	–	–	✓	–	–	–	✓	–	–	✓	–	✓	–
[26]	✓	–	–	✓	–	–	✓	–	–	–	✓	–	✓	–
[18]	✓	✓	–	–	–	–	✓	–	–	–	–	–	–	✓
[22]	✓	–	–	✓	–	–	–	–	–	✓	–	–	✓	–
[19]	✓	–	–	✓	–	–	✓	–	–	–	–	–	–	✓
[21]	✓	–	–	✓	–	–	–	✓	–	–	✓	–	–	✓
[23]	–	✓	–	✓	–	–	–	–	✓	–	✓	–	–	✓
[32]	–	✓	–	✓	–	–	–	–	–	–	✓	–	–	✓
[33]	–	✓	✓	–	–	–	✓	–	–	–	✓	–	–	✓
[24]	✓	–	–	✓	–	–	–	–	✓	–	✓	–	–	✓
[27]	✓	–	–	✓	–	Agreement index	–	✓	–	–	–	–	–	✓
[28]	✓	–	–	✓	–	Delivery accuracy	–	–	–	–	✓	–	–	✓
[29]	✓	–	–	–	–	Workload, total weighted earliness and tardiness	✓	–	–	–	–	–	✓	–

Note:

Objectives – Mksp.: makespan; Tard.: tardiness; Earli.: earliness; Env.: TEC, energy cost, electricity charge; Worker: TWC, worker fatigue.

Algorithms – EA: Evolutionary algorithm, genetic algorithm, and variants; SI: Swarm intelligence including PSO, ACO, SFLA; Co-evol.: Cooperative evolution; Greedy: Iterated greedy algorithm and variants; with RL: Algorithm with RL.

Beyond bi-objective formulations, advanced research moves toward three-objective optimization frameworks that address more comprehensive industrial requirements. Typical formulations jointly optimize fuzzy makespan, total agreement index, and fuzzy TEC to incorporate factory coordination and uncertainty [27], or combine delivery accuracy with makespan and energy consumption for customer-oriented performance [28]. Other works consider makespan, total weighted earliness and tardiness, and total workload to capture production efficiency and resource utilization simultaneously [29]. These three-objective studies usually adopt sophisticated hybrid algorithms, such as cooperative shuffled frog-leaping algorithm (SFLA) with adaptive memplex cooperation and VNS [27], three-dimensional estimation of distribution algorithms enhanced with Q-learning-based self-adaptive evolution [28], and multiobjective evolutionary algorithms with multiple neighborhood local search and simulated annealing-based replacement mechanisms [29].

The literature on multiobjective production- and green-oriented DHFSP is summarized in Table 1, which categorizes representative studies by objectives, methods, and factory configurations according to their optimization complexity. Overall, this work shows a clear trajectory from basic bi-objective formulations toward more sophisticated frameworks integrating production efficiency, environmental sustainability, and diverse industrial considerations such as lot-streaming and complex temporal constraints. However, a critical limitation persists: these studies, including advanced RL-based approaches, typically treat the distributed system as a monolithic entity and lack specialized hierarchical mechanisms that simultaneously address inter-factory coordination and intra-factory scheduling requirements.

2.2. Worker-integrated DHFSP

In real-world manufacturing environments, workers are not passive resources but active, flexible, and heterogeneous agents whose performance, skills, and costs directly impact production outcomes. This recognition has led to the emergence of worker-integrated scheduling

models. Nevertheless, research that explicitly integrates worker considerations into DHFSP remains scarce, as shown in Table 2, where most studies address only isolated worker attributes rather than comprehensive integration.

Existing works consider worker integration from different perspectives, including fatigue modeling [9], economic optimization [34], skill heterogeneity [35,36], and comprehensive worker-cost integration [37]. The research trend has evolved from treating workers as simple constraints to including them as explicit optimization objectives, but integration remains fragmented across problem settings.

Algorithmic approaches in worker-integrated DHFSP are less diverse than in traditional production-focused research. Evolutionary algorithms remain the predominant choice [9,34–36,38], while swarm intelligence methods appear less frequently [37]. RL integration shows emerging potential, with Q-learning frameworks explored for worker fatigue management [9] and adaptive optimization in worker-centric environments [34].

2.3. Research gap

The preceding literature review reveals three main research gaps for sustainable human–machine collaborative manufacturing in distributed environments: (1) most DHFSP studies oversimplify workers and ignore heterogeneous skills, productivity, and costs; (2) existing multi-objective optimization algorithms struggle to achieve both convergence and uniform solution distribution on the Pareto front in complex multi-resource settings; and (3) current RL applications are largely limited to generic parameter control, with problem specific local search for worker–machine interactions remaining underexplored.

Therefore, this study proposes a comprehensive DHFSP-WME model and a novel HMOPSO-QLS algorithm that employ multidirectional particle swarm decomposition and Q-learning-driven adaptive local search to explicitly incorporate workers and enhance Pareto front exploration in distributed human–machine collaborative manufacturing.

Table 2
Literature on multiobjective worker-integrated DHFSP.

Ref.	Objectives						Constraints	Algorithms			Factory	
	Mksp.	Tard.	Earli.	Env.	Worker	Others		EA	SI	with RL	Homo.	Hetero.
[9]	✓	–	–	✓	–	–	Worker fatigue	✓	–	✓	✓	–
[38]	–	✓	–	✓	–	–	Assembly shop	✓	–	–	✓	–
[34]	–	✓	–	✓	–	Total production cost	Time-of-use electricity price, labor shift, salary and bonus of workers	✓	–	✓	–	✓
[35]	✓	–	–	–	Imbalance degree of worker workloads	–	Multiskilled worker, transportation time	✓	–	–	–	✓
[36]	✓	–	–	–	Variance of technicians' workload	–	Fuzzy	✓	–	–	–	✓
[37]	✓	–	–	✓	Worker fatigue	Total cost	Worker and transportation factors	–	✓	–	–	✓
Ours	✓	–	–	✓	TWC	–	Worker skills, worker costs	–	✓	✓	–	✓

Note:

Objectives – Mksp.: makespan; Tard.: tardiness; Earli.: earliness; Env.: TEC, total carbon emission; Worker: worker-related objectives.

Constraints – Key resource constraints considered in the optimization model.

Algorithms – EA: Evolutionary algorithm, genetic algorithm, and variants; SI: Swarm intelligence including PSO, ACO; with RL: Algorithm with RL.

3. Problem formulation

In DHHFSP-WME, FN geographically dispersed heterogeneous factories are modeled as hybrid flow-shops with SN serial stages. At stage k of factory f , there are $MN_{f,k}$ parallel machines $M_{f,k,i}$ with possibly different processing capabilities. Each machine requires one worker. In factory f , the number of workers equals the number of machines, and workers cannot move between factories. Within a factory, each worker is permanently assigned to one machine and may operate it at any stage; workers differ in productivity and labor cost [39]. There are JN jobs. Each job is assigned to exactly one factory, must pass through all SN stages there without inter-factory transfer, and at each stage k its operation $O_{j,k}$ can be processed on any machine of that stage with standard processing time $ST_{j,k}$. The notation used in the mathematical model is summarized in Table 3.

To simultaneously minimize the makespan, TEC, and TWC, the mixed-integer linear programming model of DHHFSP-WME is formulated as follows.

$$\min. \{C_{\max}, E_{\text{total}}, WC_{\text{total}}\} \quad (1)$$

$$C_{\max} = \max_f C_{j,SN,f}, \forall j, f \quad (2)$$

$$E_{\text{total}} = \sum_{f=1}^{FN} \sum_{k=1}^{SN} \sum_{i=1}^{MN_{f,k}} (MEC_{f,k,i}^P + MEC_{f,k,i}^S), \forall f, k, i \quad (3)$$

$$WC_{\text{total}} = \sum_{f=1}^{FN} \sum_{w=1}^{MN_f} \sum_{k=1}^{SN} \sum_{i=1}^{MN_{f,k}} WC_{f,w,k,i}^P, \forall f, w, k, i \quad (4)$$

$$\text{s.t.} \sum_{f=1}^{FN} u_{j,f} = 1, \forall j \quad (5)$$

$$\sum_{i=1}^{MN_{f,k}} v_{j,f,k,i} = 1, \forall j, f, k \quad (6)$$

$$S_{j,k} \geq F_{j,k-1}, \forall j, \forall k \in \{2, 3, \dots, SN\} \quad (7)$$

$$S_{j,k} \geq 0, \forall j, k \quad (8)$$

$$F_{j,k} = S_{j,k} + AT_{j,k}, \forall j, k \quad (9)$$

$$AT_{j,k} = ST_{j,k} / (MS_{f,k,i} \times WE_{f,w,k,i}), \forall j, f, k, i, w \quad (10)$$

$$C_{j,k,f} = \begin{cases} \sum_{f=1}^{FN} \sum_{i=1}^{MN_{f,k}} v_{j,f,k,i} \times AT_{j,k}, \forall j, k = 1 \\ AT_{j,k-1} + \sum_{f=1}^{FN} \sum_{i=1}^{MN_{f,k}} v_{j,f,k,i} \times AT_{j,k}, \forall j, k > 1 \end{cases} \quad (11)$$

$$\sum_{w=1}^{MN_f} x_{f,k,i,w} = 1, \forall f, k, i \quad (12)$$

$$x_{f,k,i,w} = 0, \forall f, k, i, w \notin \{1, 2, \dots, MN_f\} \quad (13)$$

$$y_{f,k,j,j'} + y_{f,k,j',j} = 1, \forall f, k, j, j' \quad (14)$$

$$TST_{f,k,i} = \max_j (C_{j,k,f} \times v_{j,f,k,i}) - \min_j (S_{j,k} \times v_{j,f,k,i}) - \sum_{j=1}^{JN} v_{j,f,k,i} \times AT_{j,k}, \forall f, k, i \quad (15)$$

$$MEC_{f,k,i}^P = \sum_{j=1}^{JN} \sum_{w=1}^{MN_f} v_{j,f,k,i} \times x_{f,k,i,w} \times MEC_{f,k,i}^{PU} \times AT_{j,k}, \forall f, k, w \quad (16)$$

$$MEC_{f,k,i}^S = MEC_{f,k,i}^{SU} \times TST_{f,k,i}, \forall f, k, i \quad (17)$$

$$WC_{f,w,k,i}^P = (\max_j (C_{j,k,f} \times v_{j,f,k,i}) - \min_j (S_{j,k} \times v_{j,f,k,i})) \times WC_{f,w,k,i}^{PU}, \forall f, w, k, i \quad (18)$$

$$u_{j,f} \in \{0, 1\}, \forall j, f \quad (19)$$

$$v_{j,f,k,i} \in \{0, 1\}, \forall j, f, k, i \quad (20)$$

$$x_{f,k,i,w} \in \{0, 1\}, \forall f, k, i, w \quad (21)$$

$$y_{f,k,j,j'} \in \{0, 1\}, \forall f, k, j, j' \quad (22)$$

Eq. (1) defines the three optimization objectives of DHHFSP-WME, while (2)–(4) specify the makespan, TEC, and TWC, respectively. Constraints (5)–(6) ensure that each job is assigned to exactly one factory and processed by exactly one machine at each stage. Constraints (7)–(9) enforce the technological precedence of stages and define the start and completion times of each operation, and (10)–(11) determine the actual processing time and completion time of each job at each stage. Constraints (12)–(14) describe the worker-machine assignment and the processing sequence of jobs on each machine. Constraint (15) computes the total standby time of each machine, and (16)–(17) calculate the energy consumption of machines in processing and standby states, respectively. Constraint (18) determines the total labor cost of each worker. Finally, (19)–(22) define the binary decision variables.

Table 3
Notations and descriptions for the DHHFSP-WME model.

Indices	Description
f	Index of factory, ($f = 1, 2, \dots, FN$);
j, j'	Index of job, ($j, j' = 1, 2, \dots, JN$);
k	Index of stage, ($k = 1, 2, \dots, SN$);
w	Index of worker, ($w = 1, 2, \dots, MN_f$);
i	Index of machine, ($i = 1, 2, \dots, MN_{f,k}$).
Parameters	Description
FN	Total number of factories;
JN	Total number of jobs;
SN	Number of processing stages for each job;
$O_{j,k}$	Operation of job j at stage k ;
MN_f	Total number of machines in factory f ;
$MN_{f,k}$	Number of machines at stage k in factory f ;
$MS_{f,k,i}$	Standard processing speed of machine i at stage k in factory f ;
$MEC_{f,k,i}^{PU}$	Energy consumption per unit time of machine i at stage k in factory f during processing;
$MEC_{f,k,i}^{SU}$	Energy consumption per unit time of machine i at stage k in factory f during standby;
$WE_{f,w,k,i}$	Operating efficiency of worker w on machine i at stage k in factory f ;
$WC_{f,w,k,i}^{PU}$	Unit time cost of worker w operating on machine i at stage k in factory f .
Variables	Description
$MEC_{f,k,i}^P$	Energy consumption of machine i at stage k in factory f during processing;
$MEC_{f,k,i}^S$	Energy consumption of machine i at stage k in factory f during standby;
$TST_{f,k,i}$	Standby time of machine i at stage k in factory f ;
$ST_{j,k}$	Standard processing time of job j at stage k ;
$AT_{j,k}$	Actual processing time of job j at stage k ;
$S_{j,k}$	Starting time of operation $O_{j,k}$;
$F_{j,k}$	Finishing time of operation $O_{j,k}$;
$C_{j,k,f}$	Completion time of job j at stage k in factory f ;
$WC_{f,w,k,i}^P$	Total cost of worker w operating on machine i at stage k in factory f ;
$C_{\max}$	Maximum completion time of all jobs (makespan);
E_{total}	TEC;
WC_{total}	TWC.
Decision variables	Description
$u_{j,f}$	1 if job j is assigned to factory f , 0 otherwise;
$v_{j,f,k,i}$	1 if operation $O_{j,k}$ is processed on machine i in factory f , 0 otherwise;
$x_{f,k,i,w}$	1 if machine i at stage k in factory f is operated by worker w , 0 otherwise;
$y_{f,k,j,j'}$	1 if job j precedes job j' at stage k in factory f , 0 otherwise.

4. The proposed HMOPSO-QLS

4.1. Overall framework

To effectively address the DHHFSP-WME, this paper introduces HMOPSO-QLS, which combines multi-directional particle swarm optimization with Q-learning-driven local search in a unified hybrid framework. The algorithm alternates between (1) a multi-directional global search that decomposes the swarm into specialized sub-swarms to explore different regions of the objective space and (2) a two-level local search, where an inter-factory procedure redistributes jobs among factories to improve load balance and an intra-factory procedure, guided by Q-learning, adaptively selects neighborhood operators to refine schedules within each factory. After each iteration, an environmental selection mechanism based on the Pareto dominating and dominated fitness function (PDDR-FF) metric updates the swarm and archives elite solutions until the termination criterion is satisfied, yielding an approximation of the Pareto front (PF).

The algorithm framework of HMOPSO-QLS is illustrated in Fig. 1, which presents how these global and local components interact within an iterative process. The complete procedure of HMOPSO-QLS is summarized in Algorithm 1, and its components are detailed in the subsequent sections.

4.2. Encoding and decoding

The proposed algorithm employs a four-dimensional encoding scheme to represent solutions in the DHHFSP-WME, where each particle consists of four integer vectors corresponding to different decision variables:

Table 4

Example data of jobs.

Jobs ($ST_{j,k}$)	J_1	J_2	J_3	J_4	J_5
Stage 1	8	3	7	6	4
Stage 2	4	5	4	3	3

- (1) Job sequence vector (JS, v_1): represents the processing order of jobs as a non-repetitive integer permutation in the range $[1, JN]$.
- (2) Factory assignment vector (FA, v_2): encodes the factory assigned to each job as an integer sequence in the range $[1, FN]$, ensuring that each factory is allocated at least one job to maintain feasibility.
- (3) Machine assignment vector (MA, v_3): for each job, encodes the selected machine from the available machines in the first stage of its assigned factory, with values in $[1, MN_{f,1}]$.
- (4) Worker assignment vector (WA, v_4): the vector length equals the total number of machines; within each factory, it assigns workers uniquely to machines, following the constraint that each machine is operated by exactly one worker.

To illustrate the encoding and decoding method, a small-scale DHHFSP-WME example is considered with 2 factories and 5 jobs, each requiring 2 processing stages. Factory 1 has 2 machines in stage 1 and 1 machine in stage 2, whereas Factory 2 has 1 machine in stage 1 and 2 machines in stage 2. Each factory has 3 workers operating the machines. The standard processing time for each job and the parameters of machines and workers are given in Tables 4 and 5.

Note that only the assignments related to the first stage are explicitly encoded. For subsequent stages, the corresponding schedules are

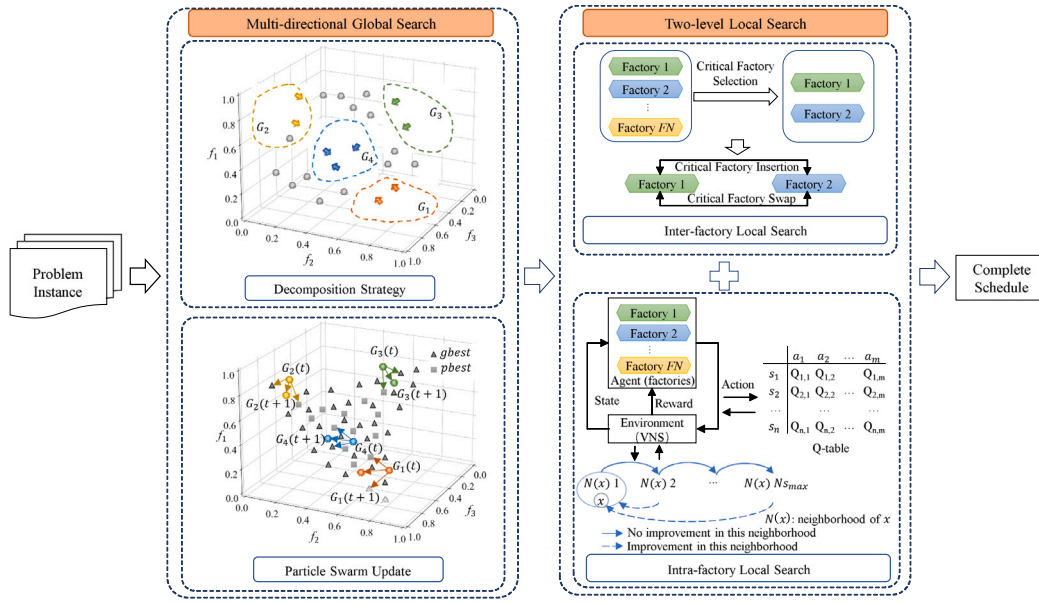


Fig. 1. The hybrid framework of HMOPSO-QLS, illustrating the global search, inter-factory local search, and Q-learning-driven intra-factory local search.

Table 5

Example data of factories, machines, and workers.

Machines($MS_{f,k,i} MEC^p_{f,k,i}$)	Factory 1			Factory 2		
	M_1		M_2	M_1		M_2
Stage 1	1 7		1.2 10	1 7		–
Stage 2	1.2 6		–	1 6		1.2 8
Workers($WE_{f,w,k,i} WC_{f,w,k,i}$)	W_1	W_2	W_3	W_1	W_2	W_3
	1.2 12	0.8 8	1.0 10	1.1 11	0.9 9	1.3 13

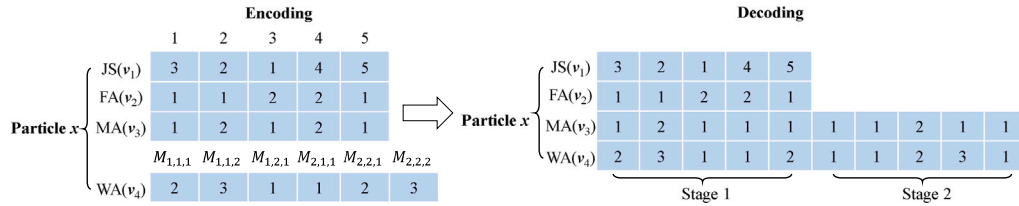


Fig. 2. Example of encoding and decoding routines.

generated according to three classical rules: the first in first out (FIFO) rule [40], the earliest completion time (ECT) rule [41], and the first available machine (FAM) rule [42].

Based on the above encoding rules, a four-vector decoding mechanism is designed to reconstruct a complete process sequence from an encoded particle, as summarized in Algorithm 2. Fig. 2 provides an example of the encoding and decoding process, showing how the four-vector representation is transformed into a feasible schedule.

The decoding process follows a stage-by-stage construction. For stage 1, jobs are directly assigned to factories, machines, and workers according to the FA, MA, and WA vectors and the job order in JS. For stages $k > 1$, the ECT rule determines the processing order of jobs based on completion times from stage $k - 1$, ties are resolved using the FIFO rule, and the FAM rule allocates jobs to machines.

It is noteworthy that this study designs a right-shifting strategy based on the characteristics of the problem [21]. By delaying the startup time of certain machines without changing C_{max} , the strategy reduces machine idle time and thus decreases both E_{total} and WC_{total} , as illustrated in Fig. 3.

4.3. Multi-directional global search

The global search phase is driven by a hybrid multiobjective particle swarm optimization (HMOPSO) algorithm. Unlike traditional PSO, this approach integrates particle swarm decomposition and specialized update operators to better balance exploration and exploitation in the multiobjective landscape.

4.3.1. Particle swarm decomposition via mixed sampling strategy

A cornerstone of the global search is particle swarm decomposition. The swarm is divided into four distinct sub-swarms (G_1, G_2, G_3, G_4) to ensure comprehensive exploration of both boundary and central regions of the PF.

This division is achieved through a mixed sampling strategy comprising two complementary components. The first component, a boundary-region sampling strategy inspired by the vector evaluated genetic algorithm (VEGA), identifies particles with the best performance on each individual objective (makespan, TEC, and TWC) to form sub-swarms G_1, G_2 , and G_3 , respectively. These elite particles guide the search toward the edges of the PF and maintain diversity along each objective dimension.

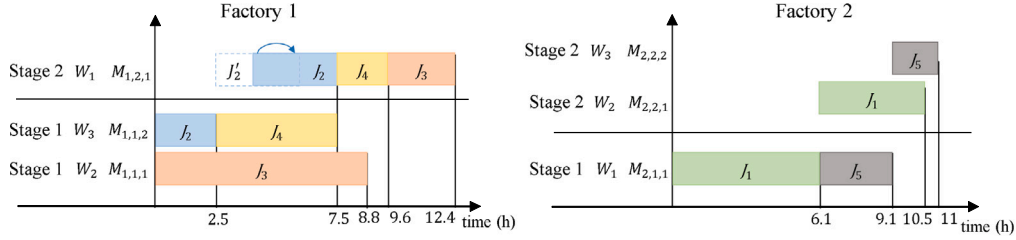


Fig. 3. Gantt chart of example schedule.

Algorithm 1 Hybrid framework of the proposed HMOPSO-QLS

Input: Data set, particle swarm size N , maximum number of evaluations E_{max} , Q-learning parameters, other parameters.
Output: Final $gbestset$ (approximated PF).
1: Set iteration counter $t \leftarrow 0$, evaluation counter $E_{cur} \leftarrow 0$.
2: Initialize particle swarm P_t with N random particles (four-vector encoding).
3: Initialize Q-table for adaptive local search.
4: Evaluate each particle in P_t ; update $E_{cur} \leftarrow N$.
5: Initialize $pbestset \leftarrow P_t$.
6: Initialize $gbestset$ with non-dominated solutions from P_t .
7: **while** $E_{cur} < E_{max}$ **do**
8: Divide P_t into four sub-swarms G_1, G_2, G_3, G_4 according to their objective values.
9: Let $P_t^G \leftarrow \emptyset$.
10: **for** each sub-swarm $G_k \in \{G_1, G_2, G_3, G_4\}$ **do**
11: **for** each particle $x_i \in G_k$ **do**
12: Update velocity and position of x_i using the update rule of G_k to obtain x_i^G .
13: Add x_i^G to P_t^G .
14: **end for**
15: **end for**
16: Evaluate P_t^G ; update $E_{cur} \leftarrow E_{cur} + N$.
17: Let $P_t^{IF} \leftarrow \emptyset$.
18: **for** each particle $x_i^G \in P_t^G$ **do**
19: Apply inter-factory local search on x_i^G to obtain x_i^{IF} .
20: Add x_i^{IF} to P_t^{IF} .
21: **end for**
22: Update E_{cur} with evaluations consumed by inter-factory local search.
23: Let $P_t^{IN} \leftarrow \emptyset$.
24: **for** each particle $x_i^{IF} \in P_t^{IF}$ **do**
25: Apply Q-learning-enhanced intra-factory local search on x_i^{IF} to obtain x_i^{IN} .
26: Update Q-table according to the reward obtained.
27: Add x_i^{IN} to P_t^{IN} .
28: **end for**
29: Update E_{cur} with evaluations consumed by intra-factory local search.
30: Let $R_t \leftarrow P_t \cup P_t^{IN}$.
31: Select the best N particles from R_t using the PDDR-FF metric to form P_{t+1} .
32: Update $pbestset$ by comparing particles in P_{t+1} with the current $pbestset$.
33: Update $gbestset$ by archiving non-dominated solutions from $pbestset$.
34: $t \leftarrow t + 1$.
35: **end while**
36: **return** $gbestset$.

Algorithm 2 Decoding method

Input: Four-vector encoded solution (JS, FA, WA, MA) .
Output: Complete feasible schedule with objective values $(C_{max}, E_{total}, WC_{total})$.
1: **for** each factory $f = 1, \dots, FN$ **do**
2: **for** each stage $k = 1, \dots, SN$ **do**
3: **if** $k = 1$ **then**
4: Assign factories, workers, and machines for the first stage according to JS , FA , WA , and MA ;
5: **else**
6: Sort completion times $C_{j,k-1,f}$ of all jobs from stage $k-1$ and determine the processing order using the ECT rule;
7: **if** multiple jobs are completed simultaneously in stage $k-1$ **then**
8: Resolve their order in stage k using the FIFO rule;
9: **end if**
10: Allocate jobs to machines using the FAM rule;
11: **end if**
12: **end for**
13: Apply the right-shifting strategy to reduce energy consumption without affecting C_{max} ;
14: **end for**
15: Compute objective values $(C_{max}, E_{total}, WC_{total})$ for the complete schedule;
16: **return** The complete schedule and $(C_{max}, E_{total}, WC_{total})$.

The second component, a central-region sampling strategy based on the PDDR-FF, selects individuals with strong dominance capability to form sub-swarm G_4 , promoting convergence toward well-balanced solutions in the central region of the PF. The selection is based on the PDDR-FF value, which evaluates an individual's dominance capability [43]:

$$eval(X_i) = |q(X_i)| + \frac{1}{|p(X_i)| + 1}, \quad i = 1, 2, \dots, popSize \quad (23)$$

where $|p(X_i)|$ is the number of individuals dominated by X_i , and $|q(X_i)|$ is the number of individuals that dominate X_i . A smaller $eval(X_i)$ indicates a stronger dominance capability. Particles with the lowest $eval$ values are selected to form the fourth sub-swarm G_4 .

Fig. 4 illustrates the mixed sampling strategy. For visualization, the three-dimensional objective space is projected onto three two-dimensional planes, as shown in Fig. 4(d)–(f). The decomposition ensures that boundary sub-swarms focus on individual objectives, while the central sub-swarm concentrates on well-balanced trade-off solutions.

4.3.2. Hybrid particle update mechanism

According to the four-vector integer encoding method described in Section 4.1, the proposed HMOPSO adopts a tailored update mechanism for discrete particles. Since changes in the JS vector have a greater impact on the schedule, an exchange-sequence operator is used to update JS by exploiting differences between two individuals and driving inferior particles toward better ones. For the other three vectors (FA, MA, and WA), a combination of crossover and mutation is adopted.

The update formulas are:

$$K = c_{v_2} \otimes m_{v_2} \otimes c_{v_3} \otimes m_{v_3} \otimes c_{v_4} \otimes m_{v_4}, \quad (24)$$

$$x(t+1) = (x(t) \oplus rs(t)) + (c_1 r_1 (pbest - x(t))) \otimes K + (c_2 r_2 (gbest - x(t))) \otimes K, \quad (25)$$

where differences between particles are represented by exchange sequences, and the number of exchange pairs is fixed, so $c_1 = c_2 = 1$. The specific values of r_1 and r_2 are given in Section 5. $rs(t)$ denotes a randomly generated exchange sequence that acts as the initial velocity of particles in the search process. Considering that changes in different vectors produce different disturbance amplitudes, different crossover and mutation rates are set for FA, MA, and WA: c_{v_2} , c_{v_3} , and c_{v_4} denote crossover rates, while m_{v_2} , m_{v_3} , and m_{v_4} denote mutation rates for FA, MA, and WA, respectively.

The particle update procedure is implemented in three steps: (1) each particle is assigned a random initial velocity for neighborhood exploration around its current position; (2) the particle is guided toward its personal best position $pbest$; and (3) the particle is further adjusted toward the global best position $gbest$. The overall process is given in Algorithm 3.

Fig. 5 illustrates the exchange-sequence operation on JS. The difference between the JS vectors of particles x and y can be represented by exchange pairs (1,2), (4,5), and (6,8). Assuming $r_1 = 0.4$, the formula selects pair (1,2) for operation, and swapping the corresponding positions yields a new JS vector.

For the remaining three vectors, a random two-point crossover strategy is used, combined with random mutation. The detailed procedure is summarized in Algorithm 4.

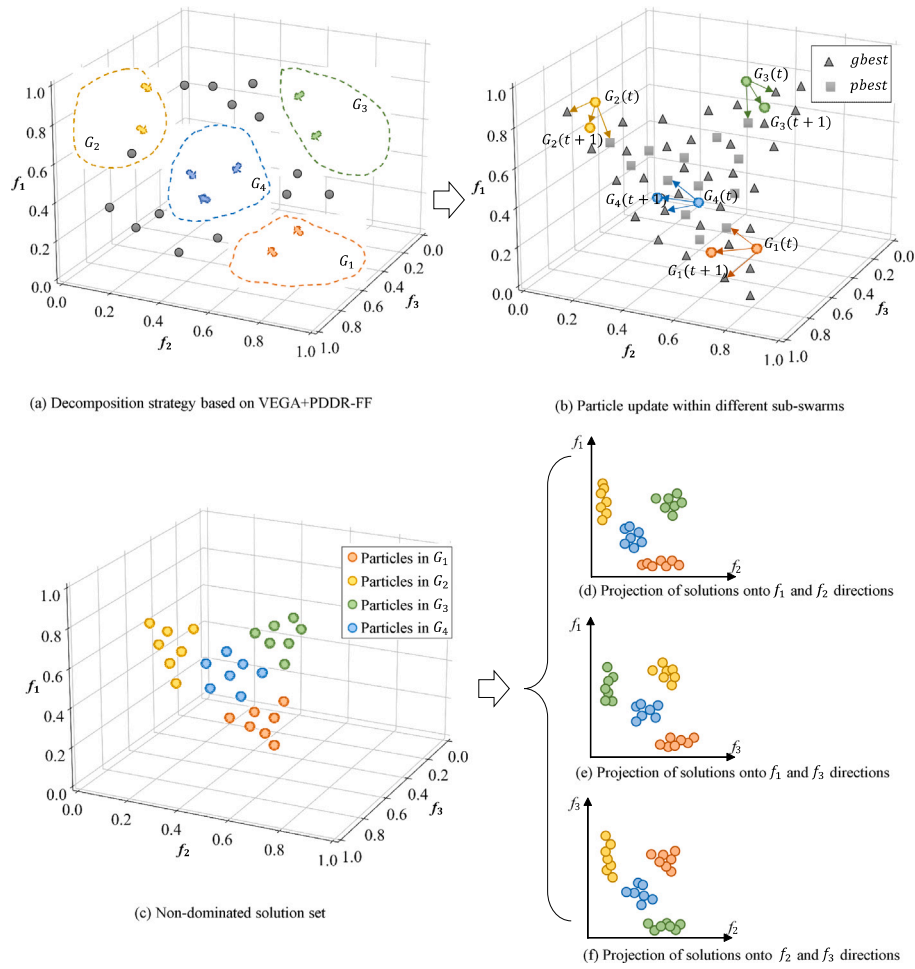


Fig. 4. Particle swarm decomposition method based on the mixed sampling strategy.

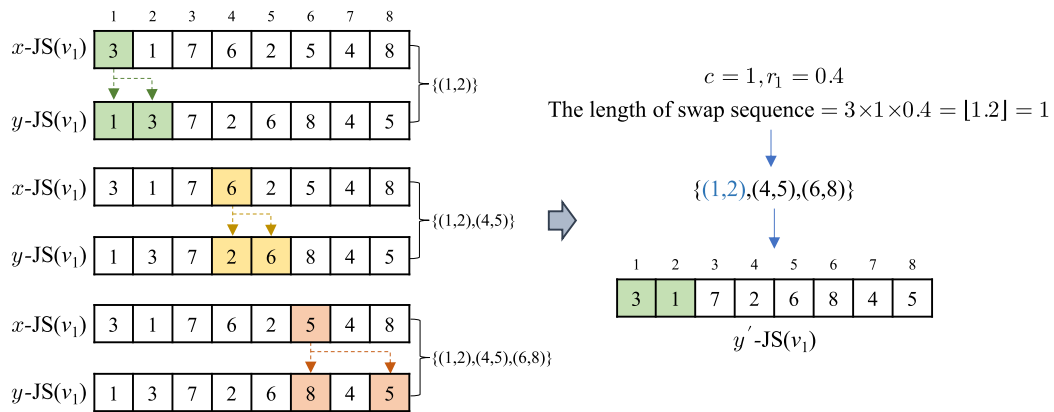


Fig. 5. Exchange sequence operation of JS.

4.4. Domain knowledge-driven inter-factory local search

After the global exploration phase, a domain knowledge-driven inter-factory local search is applied to further refine solutions. This procedure is tailored to the characteristics of each sub-swarm and exploits the notion of critical factories to achieve cross-factory load balancing, as summarized in Algorithm 5.

The domain knowledge used in this local search is centered on the concept of *critical factories*. For a given solution and a specific

objective (makespan, TEC, or TWC), the factory with the highest value of that objective is defined as the maximum critical factory (F_{max}), and the one with the lowest value is the minimum critical factory (F_{min}). Based on F_{max} and F_{min} , two operators are designed to reallocate jobs between factories and improve load balance: (1) **Critical factory swap**: randomly select one job from F_{max} and one from F_{min} and exchange their factory assignments; (2) **Critical factory insert**: randomly select a job from F_{max} and insert it into a random position in the job sequence of F_{min} .

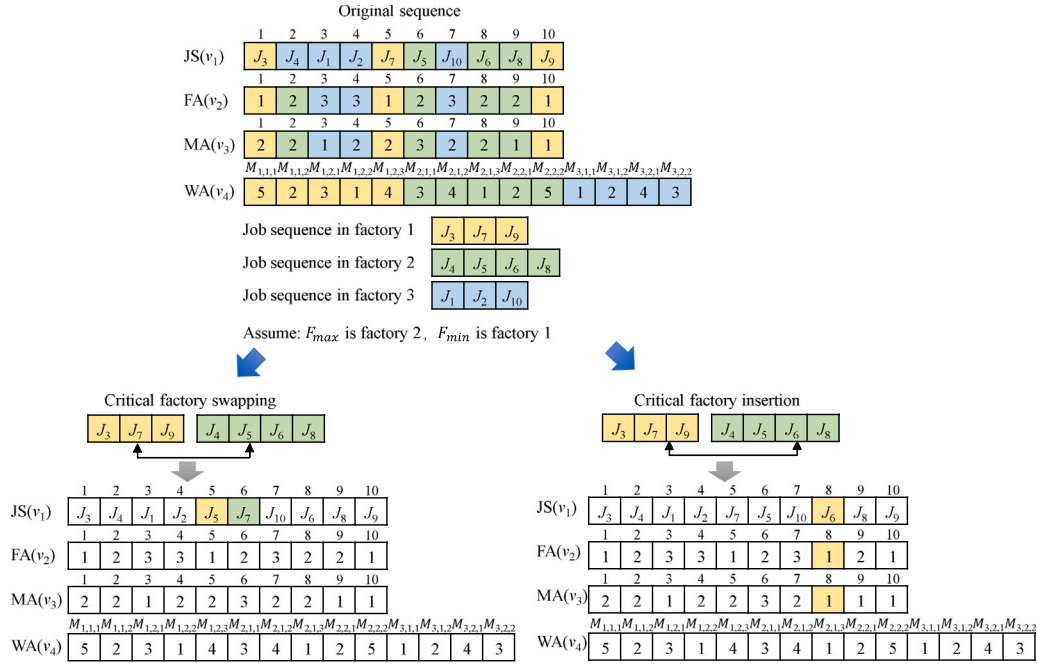


Fig. 6. Domain knowledge-driven operators for inter-factory local search.

Algorithm 3 Global search in HMOPSO-QLS

Input: Particle swarm $\{x_1(t), x_2(t), \dots, x_n(t)\}$, personal best positions $\{pbest_1, pbest_2, \dots, pbest_n\}$, global best position $gbest$, crossover rates $c_{v_2}, c_{v_3}, c_{v_4}$, mutation rates $m_{v_2}, m_{v_3}, m_{v_4}$.

Output: Updated particle swarm $\{x_1(t+1), x_2(t+1), \dots, x_n(t+1)\}$.

- 1: for $i = 1$ to n do
- 2: Generate random exchange sequence $rs_i(t)$ as initial velocity;
- 3: $x_i^{(1)}(t) \leftarrow$ update $x_i(t)$ through neighborhood search using $rs_i(t)$;
- 4: Compute difference $\Delta_{pb} = x_i^{(1)}(t) - pbest_i$;
- 5: $x_i^{(2)}(t) \leftarrow$ update JS vector of $x_i^{(1)}(t)$ using exchange sequence based on Δ_{pb} ;
- 6: $x_i^{(3)}(t) \leftarrow$ perform crossover and mutation on FA, MA, and WA between $x_i^{(2)}(t)$ and $pbest_i$;
- 7: Compute difference $\Delta_{gb} = x_i^{(3)}(t) - gbest$;
- 8: $x_i^{(4)}(t) \leftarrow$ update JS vector of $x_i^{(3)}(t)$ using exchange sequence based on Δ_{gb} ;
- 9: $x_i(t+1) \leftarrow$ perform crossover and mutation on FA, MA, and WA between $x_i^{(4)}(t)$ and $gbest$;
- 10: end for
- 11: return $\{x_1(t+1), x_2(t+1), \dots, x_n(t+1)\}$.

Fig. 6 shows the inter-factory local search in detail. For particles in the boundary-region sub-swarms (G_1, G_2, G_3), the inter-factory local search is guided by the corresponding single objective: for example, particles in G_1 (focused on makespan) accept a move only if it improves makespan, i.e., $f_1(x') < f_1(x)$. For particles in the central-region sub-swarm (G_4), the objective is to obtain better trade-off solutions, and a move is accepted only if the new solution x' has a lower PDDR-FF value, i.e., $eval(x') < eval(x)$.

The two operators are applied sequentially with a restart mechanism upon successful improvement, ensuring efficient refinement. This strategy enhances cross-factory load balancing through targeted job reallocation and mitigates performance bottlenecks caused by uneven resource distribution.

4.5. Q-learning enhanced intra-factory local search

Building upon the inter-factory local search that redistributes workload among factories, an intra-factory local search strategy is further introduced to refine schedules within each factory. To adaptively decide which factory to intensify and how to search within it, a Q-learning-based VNS strategy is proposed.

Algorithm 4 Crossover and mutation process in HMOPSO-QLS

Require: Particle swarm $\{x_1(t), x_2(t), \dots, x_n(t)\}$, personal best positions $\{pbest_1, pbest_2, \dots, pbest_n\}$, global best position $gbest$, crossover rates $\{c_{v_2}, c_{v_3}, c_{v_4}\}$, mutation rates $\{m_{v_2}, m_{v_3}, m_{v_4}\}$.

Ensure: Updated particle swarm $\{x_1^*(t), x_2^*(t), \dots, x_n^*(t)\}$.

- 1: for each particle $x_i(t)$ do
- 2: for $parent \in \{pbest_i, gbest\}$ do
- 3: Generate $r_1, \dots, r_6 \sim U(0, 1)$;
- 4: if $r_1 < c_{v_2}$ then
- 5: Crossover FA vectors of $x_i(t)$ and $parent$ at the same positions;
- 6: if resulting machine index in MA is infeasible then
- 7: Replace with a random machine from the current stage;
- 8: end if
- 9: end if
- 10: if $r_2 < c_{v_3}$ then
- 11: Crossover MA vectors of $x_i(t)$ and $parent$ at the same positions;
- 12: if machine index does not belong to the assigned factory then
- 13: Replace with a random machine from the current stage;
- 14: end if
- 15: end if
- 16: if $r_3 < c_{v_4}$ then
- 17: Crossover WA vectors of $x_i(t)$ and $parent$ within the same factory;
- 18: end if
- 19: if $r_4 < m_{v_2}$ then
- 20: Mutate a random position in the FA vector of $x_i(t)$;
- 21: end if
- 22: if $r_5 < m_{v_3}$ then
- 23: Mutate a random position in the MA vector of $x_i(t)$;
- 24: end if
- 25: if $r_6 < m_{v_4}$ then
- 26: Perform adjacent-swap mutation on the WA vector of a random factory;
- 27: end if
- 28: end for
- 29: end for
- 30: return $\{x_1^*(t), x_2^*(t), \dots, x_n^*(t)\}$;

4.5.1. Q-learning-based adaptive factory selection strategy

Q-learning, a model-free RL algorithm, is employed to learn an optimal policy for selecting which factory to search. It estimates the long-term value of taking an action (selecting a factory) in a given state and updates the Q-values by the Bellman equation:

$$Q(s_t, a_t) \leftarrow Q(s_t, a_t) + \alpha \left(R(s_t, a_t) + \gamma \max_a Q(s_{t+1}, a) - Q(s_t, a_t) \right), \quad (26)$$

Algorithm 5 Domain knowledge-driven inter-factory local search

Input: Four sub-swarms (G_1, G_2, G_3, G_4) after global search.
Output: Updated sub-swarms after local search.

```

1: for each particle  $x$  in the population do
2:   Initialize operator index  $t = 1$ ;
3:   while  $t \leq 2$  do
4:     if  $t = 1$  then
5:       Generate a new solution  $x' = \text{CriticalFactorySwap}(x)$ ;
6:     else
7:       Generate a new solution  $x' = \text{CriticalFactoryInsert}(x)$ ;
8:     end if
9:     accepted  $\leftarrow$  false;
10:    if  $x \in G_i$  where  $i \in \{1, 2, 3\}$  then
11:      if  $f_i(x') < f_i(x)$  then
12:         $x \leftarrow x'$ ; accepted  $\leftarrow$  true;
13:      end if
14:    else if  $x \in G_4$  then
15:      if  $\text{eval}(x') < \text{eval}(x)$  then
16:         $x \leftarrow x'$ ; accepted  $\leftarrow$  true;
17:      end if
18:    end if
19:    if accepted then
20:       $t \leftarrow 1$ ;
21:    else
22:       $t \leftarrow t + 1$ ;
23:    end if
24:  end while
25: end for
26: return Updated sub-swarms.

```

where s_t and a_t are the current state and action, $R(s_t, a_t)$ is the immediate reward, α is the learning rate, and γ is the discount factor. By iteratively updating the Q-table, the agent gradually learns an effective factory-selection policy.

Algorithm 6 Q-learning enhanced intra-factory local search

Input: Particle swarm, action set $\mathcal{A}$, state set $\mathcal{S}$, RL parameters (α, γ, ϵ), maximum iterations I_{max} .
Output: Updated particle swarm.

```

1: for each particle  $x$  in the swarm do
2:   Initialize state  $s$  for  $x$ ; initialize Q-table  $Q(s, a)$ ;
3:   for  $i = 1$  to  $I_{max}$  do
4:     Generate a random number  $r \in (0, 1)$ ;
5:     if  $r < \epsilon$  then
6:       Randomly select an action  $a \in \mathcal{A}$  (exploration);
7:     else
8:       Select action  $a = \arg \max_{a' \in \mathcal{A}} Q(s, a')$  (exploitation);
9:     end if
10:    Execute VNS on the factory specified by  $a$ , yielding a new solution  $x'$ ;
11:    Calculate reward  $R(s, a)$  based on the sub-swarm of  $x$  (Eqs. (27)–(30));
12:    Observe the new state  $s'$  based on the fitness change (Table 6);
13:    Update Q-table using Eq. (26);
14:     $s \leftarrow s'$ ;  $x \leftarrow x'$ ;
15:  end for
16: end for
17: return Updated particle swarm.

```

Implementing Q-learning requires defining the state space, action space, and reward function. Considering the characteristics of DHHFSP-WME and the sub-swarm structure, they are designed as follows:

- **State:** The three optimization objectives are to minimize makespan, TEC, and TWC. To reduce the state space, the historical change in a particle's fitness is used as the state signal. All particles are initially set to state 1. After selecting an action and updating a particle, the new solution is evaluated. For G_1 – G_3 , the performance change Δf is measured in the single-objective direction corresponding to each sub-swarm; for G_4 , Δf is defined as the sum of change rates across all three objectives. If $\Delta f > 0$, the state is set to 0; otherwise, it is set to 1. The resulting eight states are summarized in Table 6.
- **Action:** Since the problem involves multiple factories with different performance levels, applying VNS to all factories would be computationally expensive. Therefore, the action set is defined as

Table 6

State definitions for Q-learning.

Subgroup	State ID	Value	Condition	Calculation of Δf
G_1	S_1	0	$\Delta f > 0$	$f_1(x) - f_1(x')$
	S_2	1	$\Delta f \leq 0$	
G_2	S_3	0	$\Delta f > 0$	$f_2(x) - f_2(x')$
	S_4	1	$\Delta f \leq 0$	
G_3	S_5	0	$\Delta f > 0$	$f_3(x) - f_3(x')$
	S_6	1	$\Delta f \leq 0$	
G_4	S_7	0	$\Delta f > 0$	$\sum_{i=1}^3 \frac{f_i(x) - f_i(x')}{f_i(x)}$
	S_8	1	$\Delta f \leq 0$	

Table 7

Action definitions for Q-learning.

Action	Specific operation
a_1	Select factory 1 for VNS
a_2	Select factory 2 for VNS
a_3	Select factory 3 for VNS
a_4	Select factory 4 for VNS

factory selection: at each step, the agent chooses one factory for VNS. Assuming there are 4 factories, the actions are summarized in Table 7. For each particle, the agent selects a factory as an action, and then VNS is executed within the selected factory.

- **Reward:** Because the sub-swarms are defined according to different objective directions, the reward design also follows this decomposition. As shown in Eqs. (27)–(30), for G_1 – G_3 , the reward is the improvement in the corresponding single objective between the solution after VNS, $\text{VNS}(x)$, and the original solution x . For G_4 , the reward is defined as the sum of relative improvements across all three objectives. This design provides more informative feedback to the agent after each action.

$$R(s, a) = f_1(x) - f_1(\text{VNS}(x)), \quad \text{if } x \in G_1, \quad (27)$$

$$R(s, a) = f_2(x) - f_2(\text{VNS}(x)), \quad \text{if } x \in G_2, \quad (28)$$

$$R(s, a) = f_3(x) - f_3(\text{VNS}(x)), \quad \text{if } x \in G_3, \quad (29)$$

$$R(s, a) = \sum_{i=1}^3 \frac{f_i(x) - f_i(\text{VNS}(x))}{f_i(x)}, \quad \text{if } x \in G_4. \quad (30)$$

4.5.2. Domain knowledge-driven VNS for intra-factory optimization

Once Q-learning selects a factory, VNS is executed to explore solutions within that factory. Based on the problem characteristics, three local search operators (insert, reverse, swap) are designed for each of the three feature vectors (JS, MA, WA), yielding a diversified set of $3 \times 3 = 9$ neighborhood structures, as summarized in Table 8. For each operator, two positions are randomly selected within the selected factory for the corresponding vector, and the operator then performs insertion, reverse, or swap between these positions.

Algorithm 7 describes the VNS procedure in detail.

This Q-learning-driven VNS strategy adaptively selects factories and systematically applies neighborhood operators within the selected factory to fine-tune particles, thereby effectively enhancing solution quality.

5. Experimental results

Since there is currently no benchmark dataset for the DHHFSP-WME problem, this study augments the dataset in [21] with worker efficiency and cost constraints to generate instances of varying sizes. The number of jobs JN is set to $\{20, 50, 100, 150, 200\}$, the number of stages SN to $\{2, 5, 8\}$, and the number of factories FN to $\{3, 4, 5\}$. The unit-time work efficiency $WE_{f,w,k,i}$ of worker w at stage k for job i in factory f follows a uniform distribution $U(0.5, 1.5)$, and the corresponding unit-time cost is set to $WE_{f,w,k,i} \times 10$. One instance is generated for each combination

Table 8
Summary of VNS operators.

No.	Vector	Operator	Description
1	JS	Insert	Insert the element at the second position before the first position.
2	JS	Reverse	Reverse the subsequence between the two positions.
3	JS	Swap	Swap the elements at the two positions.
4	MA	Insert	Insert the element at the second position before the first position.
5	MA	Reverse	Reverse the subsequence between the two positions.
6	MA	Swap	Swap the elements at the two positions.
7	WA	Insert	Insert the element at the second position before the first position.
8	WA	Reverse	Reverse the subsequence between the two positions.
9	WA	Swap	Swap the elements at the two positions.

Algorithm 7 VNS for intra-factory local search

Input: A particle x , neighborhood structures $N_s(x)$ for $s = 1, \dots, s_{\max}$.
Output: An updated particle x .
1: Let $s = 1$;
2: **while** $s \leq s_{\max}$ **do**
3: **Shake:** generate x' from x using neighborhood $N_s(x)$;
4: **Local search (VND):** find the best solution x'' in the neighborhood of x' ;
5: accepted $\leftarrow$ **false**;
6: **if** $x \in G_i$ where $i \in \{1, 2, 3\}$ **then**
7: **if** $f_i(x'') < f_i(x)$ **then**
8: $x \leftarrow x''$; accepted $\leftarrow$ **true**;
9: **end if**
10: **else if** $x \in G_4$ **then**
11: **if** $\text{eval}(x'') < \text{eval}(x)$ **then**
12: $x \leftarrow x''$; accepted $\leftarrow$ **true**;
13: **end if**
14: **end if**
15: **if** accepted **then**
16: $s \leftarrow 1$;
17: **else**
18: $s \leftarrow s + 1$;
19: **end if**
20: **end while**
21: **return** An updated particle x .

of $\{JN, SN, FN\}$, resulting in $5 \times 3 \times 3 = 45$ test instances, named using the convention JN_SN_FN .

This study compares HMOPSO-QLS with MOEA-LS [29], MOPSODS-DE [44], MOHEA-DE [45], MOPSO [46], NSGA-II [47], and SPEA2 [48]. All algorithms are implemented in Java and executed on a Windows 10 computer with an Intel(R) Core(TM) i7-7700 CPU at 3.60 GHz and 16 GB RAM. Each instance is run 30 times independently, and the maximum number of function evaluations is set to 500,000 for all algorithms. To assess multiobjective performance in terms of both convergence and diversity, we adopt four standard indicators: inverted generational distance (IGD), hypervolume (HV), spacing (SP), and the coverage (C). For each instance, the non-dominated solution sets obtained by different algorithms over the 30 runs are collected and evaluated using these metrics, with the C -metric further characterizing the pairwise dominance relationships between algorithms.

5.1. Parameter settings

Since HMOPSO-QLS consists of a MOHPSO-based global search and a Q-learning-driven local search, the parameters of MOHPSO are tuned first. The main parameters include r_1 , r_2 , c_{v_2} , c_{v_3} , c_{v_4} , m_{v_2} , m_{v_3} , and m_{v_4} . A full factorial experiment is conducted on a representative instance 50_5_4 (50 jobs, 5 stages, 4 factories), with each parameter taking four candidate values and each setting independently executed 30 times. The particle swarm size is set to 100, following [21]. For swarm division, four sub-swarm size configurations are considered: $M_1 = \{10, 10, 10, 70\}$, $M_2 = \{15, 15, 15, 55\}$, $M_3 = \{20, 20, 20, 40\}$, and $M_4 = \{25, 25, 25, 25\}$, where the first three values denote the sizes of sub-swarms G_1 – G_3 and the fourth value denotes the size of G_4 .

As shown in Fig. 7, the best performance in terms of IGD and HV is obtained with $r_1 = r_2 = 0.4$, $c_{v_2} = 0.2$, $c_{v_3} = 0.4$, $c_{v_4} = 0.3$, $m_{v_2} = 0.06$, $m_{v_3} = 0.1$, $m_{v_4} = 0.25$, and the clustering method M_2 (i.e., sub-swarms G_1 – G_3 of size 15 and G_4 of size 55).

After determining the parameters for MOHPSO, the remaining parameters for HMOPSO-QLS are tuned, including the number of Q-learning iterations (Q_Times), the number of local search iterations (L_Times), the discount factor (γ), and the greedy rate (ϵ). Each parameter is assigned four candidate values, and a full factorial experiment is again performed. As illustrated in Fig. 8, the best combination is $Q_Times = 40$, $\gamma = 0.8$, $\epsilon = 0.9$, and $L_Times = 40$.

For fairness, the parameters of all comparison algorithms are also tuned using full factorial experiments under a fixed population size of 100. The final parameter configurations used in subsequent experiments are summarized in Table 9.

5.2. Effectiveness of local search

To evaluate the effectiveness of the proposed inter- and intra-factory local search strategies (Sections 4.4 and 4.5), four variants are considered:

- (1) MOHPSO: baseline without local search;
- (2) MOHPSO-LS1: MOHPSO with the inter-factory local search only;
- (3) MOHPSO-LS2: MOHPSO-LS1 further combined with a multiobjective VNS (without Q-learning) for intra-factory local search;
- (4) HMOPSO-QLS: the proposed algorithm, integrating both inter-factory local search and Q-learning-enhanced multiobjective VNS.

Tables A.11–A.13 report the IGD , HV , and SP results. Compared with MOHPSO, MOHPSO-LS1 achieves the best IGD on all 45 instances and the best HV and SP on 43 and 44 instances, respectively, confirming the benefit of the inter-factory local search. Compared with MOHPSO-LS1, MOHPSO-LS2 further improves all three indicators on most instances, demonstrating the effectiveness of the VNS strategy. Finally, HMOPSO-QLS consistently outperforms MOHPSO-LS2 on all three metrics, showing that Q-learning-guided intra-factory VNS yields additional performance gains.

In summary, the combination of critical-factory-based inter-factory local search and Q-learning-driven intra-factory multiobjective VNS effectively enhances convergence and diversity by balancing factory loads and adaptively selecting factories for intensification.

5.3. Results and analysis

This study compares the proposed HMOPSO-QLS algorithm with MOPSO, MOEA-LS, MOPSODS-DE, MOHEA-DE, NSGA-II, and SPEA2 on 45 instances of varying scales. Using the experimental setup described in Section 5, each algorithm is independently run 30 times on each instance, and performance is assessed by the IGD , HV , SP , and C metrics.

The IGD metric mainly measures convergence to the true PF (smaller values indicate better convergence). As shown in Table A.14, HMOPSO-QLS achieves significantly better IGD values than the comparison algorithms on 43 out of 45 test instances, and is slightly inferior to MOHEA-DE only on instances “50_8_5” and “100_8_5”. This result indicates that HMOPSO-QLS generally exhibits superior convergence performance.

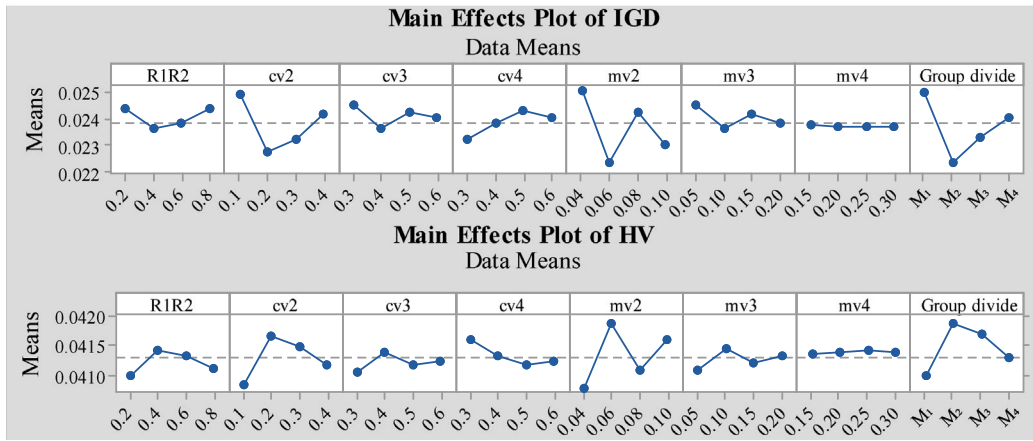
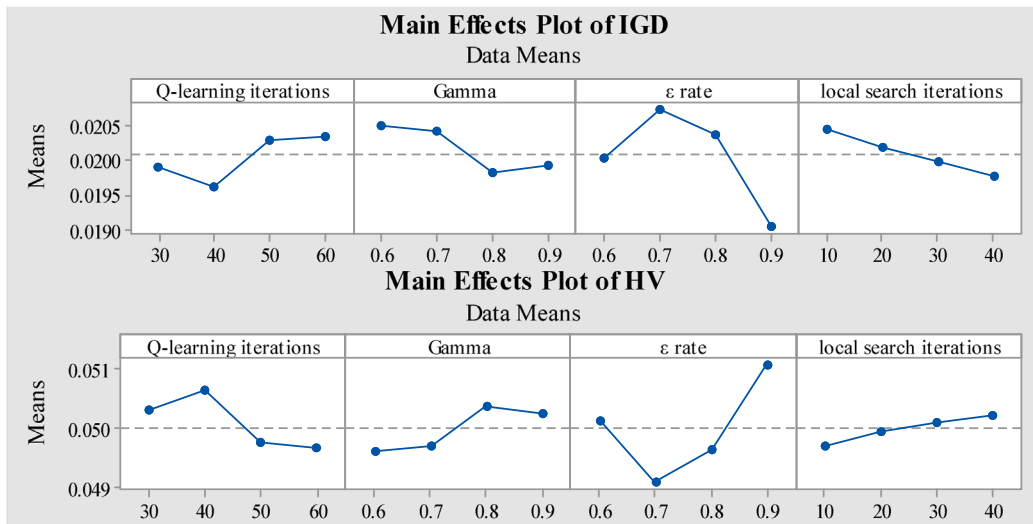
Fig. 7. Main effects on *IGD* and *HV* in MOHPSO.Fig. 8. Main effects on *IGD* and *HV* in HMOPSO-QLS.

Table 9
Parameter settings.

Parameter	MOPSODS-DE	MOEA-LS	MOHEA-DE	MOPSO	NSGA-II	SPEA2	HMOPSO-QLS
Population size	100	100	100	100	100	100	100
Elite population size	–	–	100	–	100	100	–
Cluster size	M_2	–	–	–	–	–	M_2
r_1, r_2	0.4	–	–	0.4	–	–	0.4
c_{v_1}	–	0.6	0.6	–	0.4	0.3	–
c_{v_2}	0.2	0.2	0.2	0.2	0.2	0.3	0.2
c_{v_3}	0.3	0.3	0.4	0.4	0.3	0.4	0.4
c_{v_4}	0.4	0.4	0.4	0.3	0.4	0.4	0.3
m_{v_1}	–	0.4	0.3	–	0.2	0.3	–
m_{v_2}	0.06	0.06	0.08	0.08	0.04	0.04	0.06
m_{v_3}	0.1	0.15	0.15	0.15	0.1	0.15	0.1
m_{v_4}	0.2	0.15	0.15	0.15	0.15	0.05	0.25
Neighborhood size	–	30	–	–	–	–	–
DE probability	0.8	–	0.7	–	–	–	–
Q_{Times}	–	–	–	–	–	–	40
L_{Times}	–	–	–	–	–	–	40
Discount factor γ	–	–	–	–	–	–	0.8
Greediness rate ε	–	–	–	–	–	–	0.9
Maximum evaluations	500 000	500 000	500 000	500 000	500 000	500 000	500 000

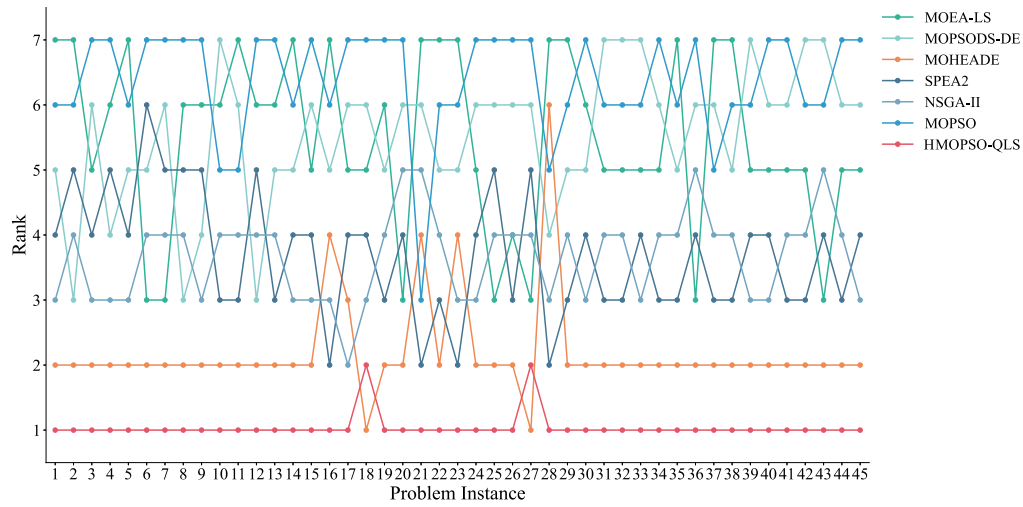


Fig. 9. Ranking of all algorithms in *IGD* index results.

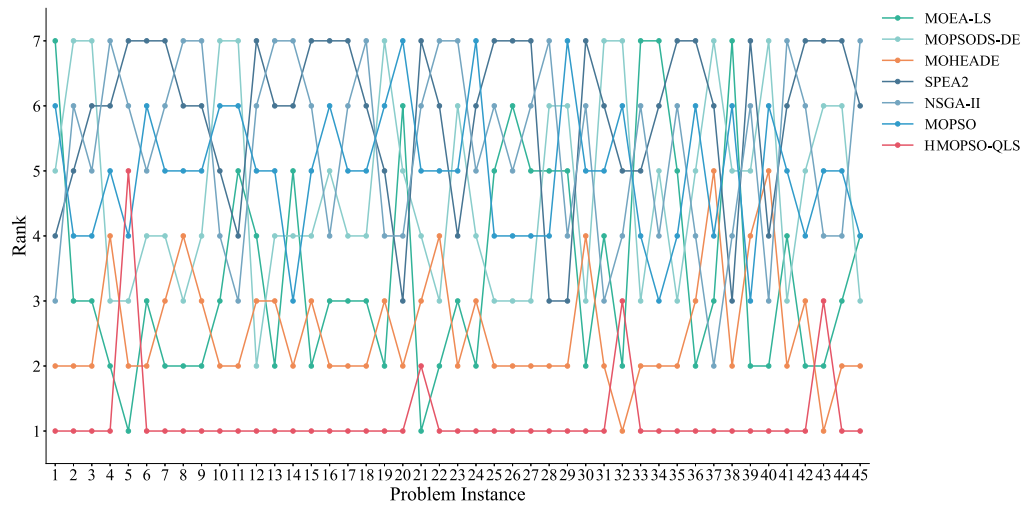


Fig. 10. Ranking of all algorithms in *HV* index results.

The *HV* metric evaluates both convergence and diversity (larger values indicate better performance). Table A.15 shows that HMOPSO-QLS attains significantly higher *HV* values than the comparison algorithms on 41 instances. Although MOHEA-DE and MOEA-LS obtain slightly higher *HV* values on four instances, the differences are not statistically pronounced. Overall, HMOPSO-QLS demonstrates excellent diversity in most of the test problems.

The *SP* metric is used to assess the uniformity of the solution set distribution (smaller values indicate more uniform spacing). According to Table A.16, HMOPSO-QLS achieves the best *SP* values on 42 out of 45 test instances, and the significance analysis confirms these improvements. These results suggest that HMOPSO-QLS provides a more uniformly distributed approximation to the PF in most cases.

To facilitate a clear visual analysis of the experimental results, the *IGD* and *Spacing* metrics are presented in ascending order, while the *HV* metric is presented in descending order, as illustrated in Figs. 9–11. The problem instance numbers in these figures correspond to the sequence listed in Table A.14. It is evident that HMOPSO-QLS outperforms the comparison algorithms in the majority of instances, achieving superior values across all three metrics.

The *C* metric describes the coverage and convergence relation between two solution sets, where a larger $C(A, B)$ value indicates stronger coverage of algorithm *A* relative to *B*. From Table A.17, it can be observed that HMOPSO-QLS consistently outperforms all other

algorithms across all problem sizes, indicating better convergence and dominance performance.

To further investigate the statistical significance of the observed differences, the Wilcoxon rank-sum test is conducted on the results of all instances and metrics. As summarized in Table 10, HMOPSO-QLS significantly outperforms the comparison algorithms in most cases, and the improvements are statistically significant at the conventional confidence levels.

In conclusion, based on the comprehensive evaluation using *IGD*, *HV*, *SP*, and *C* metrics, HMOPSO-QLS clearly outperforms the comparison algorithms in terms of both convergence and distribution. This superior performance can be attributed to the multi-directional decomposition and update strategies and the improved particle swarm update mechanism in the global search, the inter-factory local search strategy that balances workloads among factories, and the Q-learning-driven intra-factory multiobjective VNS strategy that further fine-tunes particles and enhances solution quality.

6. Limitations and future directions

This study successfully applies the HMOPSO-QLS algorithm to the DHHFSP, validating its effectiveness. However, we recognize certain limitations in both the algorithm and the mathematical model that warrant further investigation.

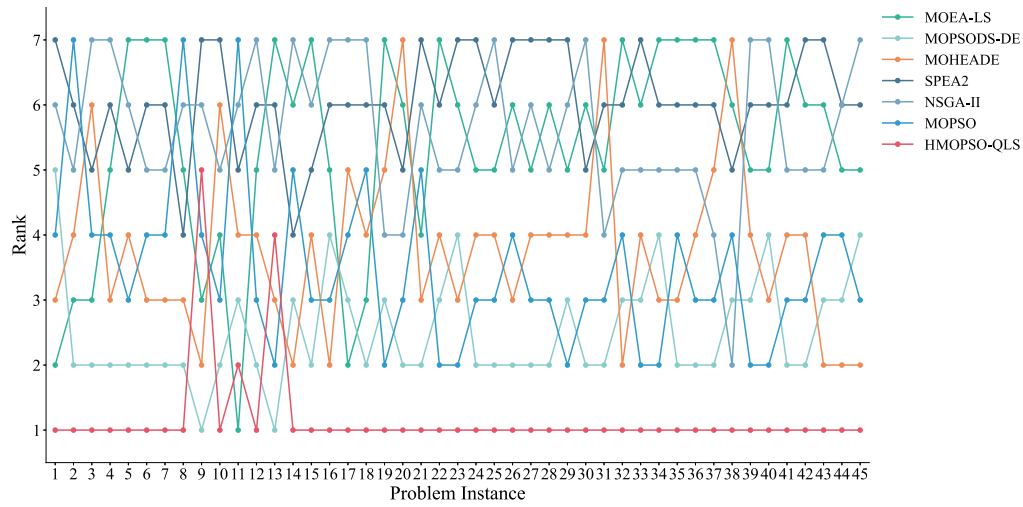


Fig. 11. Ranking of all algorithms in *Spacing* index results.

Table 10
Wilcoxon rank sum test results.

	w/t/1	R+	R-	p-value	h (alpha = 0.5)
<i>IGD</i>					
HMOPSO-QLS vs. MOEA-LS	45/0/0	1342	8	4.56E–43	1
HMOPSO-QLS vs. MOHEADE	44/0/1	1331	19	3.41E–17	1
HMOPSO-QLS vs. MOPSODS-DE	43/0/2	1319	31	2.56E–15	1
HMOPSO-QLS vs. NSGA-II	45/0/0	1340	10	7.11E–24	1
HMOPSO-QLS vs. SPEA2	45/0/0	1338	12	8.73E–23	1
HMOPSO-QLS vs. MOPSO	45/0/0	1336	14	5.76E–20	1
<i>HV</i>					
HMOPSO-QLS vs. MOEA-LS	43/0/2	1322	28	5.77E–26	1
HMOPSO-QLS vs. MOHEADE	44/0/1	1327	23	4.54E–24	1
HMOPSO-QLS vs. MOPSODS-DE	40/0/5	1289	61	3.39E–14	1
HMOPSO-QLS vs. NSGA-II	45/0/0	1343	7	3.59E–61	1
HMOPSO-QLS vs. SPEA2	45/0/0	1336	14	5.89E–38	1
HMOPSO-QLS vs. MOPSO	45/0/0	1335	15	8.98E–35	1
<i>SP</i>					
HMOPSO-QLS vs. MOEA-LS	44/0/1	1333	17	3.43E–20	1
HMOPSO-QLS vs. MOHEADE	42/1/2	1268	82	1.58E–08	1
HMOPSO-QLS vs. MOPSODS-DE	45/0/0	1340	10	3.33E–28	1
HMOPSO-QLS vs. NSGA-II	45/0/0	1332	18	6.35E–21	1
HMOPSO-QLS vs. SPEA2	45/0/0	1341	9	9.98E–32	1
HMOPSO-QLS vs. MOPSO	45/0/0	1344	6	3.56E–37	1

First, regarding the algorithm, the current study focuses on three optimization objectives. In practical scenarios, decision-makers often encounter many-objective problems (more than three objectives). The effectiveness of the current decomposition and exploration strategies in such high-dimensional spaces requires further validation. Furthermore, while the Q-learning mechanism effectively adapts to the heterogeneity of factories, the continuous updating of the Q-table incurs additional computational overhead. To address this, our future work will explore more sample-efficient RL techniques to reduce the computational cost of the local search while maintaining solution quality.

Second, regarding the scheduling model and Industry 5.0, the current model primarily focuses on productivity and labor costs. Although effective, this perspective does not fully align with the “human-centric” principle of Industry 5.0. Factors such as fatigue accumulation, emotional changes, and learning effects are not deeply integrated. Therefore, the next phase of our research will focus on comprehensive worker modeling. We aim to design “human-centered” optimization objectives that consider the physiological and psychological states of workers.

Finally, in terms of dynamic uncertainty, the current model assumes a relatively static environment. In reality, sudden disruptions such as machine failures, rush orders, fluctuating energy prices, and

unexpected worker absences can significantly impact schedule efficiency. Future research will extend the model to dynamic environments by adopting stochastic or robust optimization methods to integrate these real-world uncertainties, thereby producing schedules that are more resilient to the randomness of human behavior and production conditions.

7. Conclusion

This study investigated the DHHFSP in geographically dispersed factories with heterogeneous machines and skilled workers. To better reflect human-machine collaboration in such systems, we developed a worker-machine-environment collaborative scheduling model that jointly minimizes makespan, TEC, and TWC.

To solve this challenging multiobjective problem, we proposed the HMOPSO-QLS. HMOPSO-QLS consists of three main components: (1) a hybrid framework that combines multidirectional particle swarm-based global search with RL-based local search; (2) a Pareto front-oriented four-directional swarm decomposition and update mechanism, where three boundary-exploration sub-swarms and one central sub-swarm cooperatively guide particle evolution to improve convergence and diversity; and (3) a two-level local search scheme that links domain

Table A.11

Index results of MOHPSO and MOHPSO-LS1 for DHHFSP-WME.

Problems	IGD		HV		SP	
	MOHPSO	MOHPSO-LS1	MOHPSO	MOHPSO-LS1	MOHPSO	MOHPSO-LS1
20_2_3	1.22E-02 +	1.08E-02	4.10E-02 +	4.55E-02	1.18E-02 +	1.05E-02
20_2_4	3.03E-02 +	2.67E-02	1.23E-02 +	1.36E-02	1.26E-02 +	1.12E-02
20_2_5	5.70E-02 +	5.01E-02	2.77E-02 +	3.08E-02	8.11E-03 +	7.22E-03
20_5_3	2.86E-02 +	2.51E-02	6.04E-02 +	6.70E-02	1.04E-02 +	9.26E-03
20_5_4	3.31E-02 +	2.91E-02	2.43E-03 +	2.70E-03	9.75E-03 +	8.68E-03
20_5_5	3.76E-02 +	3.31E-02	1.18E-02 +	1.30E-02	1.28E-02 +	1.14E-02
20_8_3	1.89E-02 +	1.66E-02	8.67E-03 +	9.62E-03	9.50E-03 +	8.45E-03
20_8_4	3.60E-02 +	3.16E-02	6.80E-02 +	7.55E-02	9.99E-03 +	8.89E-03
20_8_5	5.88E-02 +	5.17E-02	8.09E-03 +	8.98E-03	2.20E-02 +	1.96E-02
50_2_3	3.48E-02 +	3.06E-02	5.36E-02 +	5.95E-02	7.51E-03 +	6.69E-03
50_2_4	1.47E-02 +	1.30E-02	1.51E-02 +	1.67E-02	1.80E-02 +	1.60E-02
50_2_5	2.69E-02 +	2.37E-02	5.01E-03 +	5.56E-03	9.65E-03 +	8.59E-03
50_5_3	2.64E-02 +	2.32E-02	6.35E-02 +	7.05E-02	7.22E-03 +	6.43E-03
50_5_4	4.39E-02 +	3.86E-02	7.30E-03 +	8.10E-03	1.25E-02 +	1.11E-02
50_5_5	3.33E-02 +	2.92E-02	1.03E-02 +	1.15E-02	7.89E-03 +	7.02E-03
50_8_3	2.75E-02 +	2.42E-02	9.00E-03 +	9.99E-03	6.37E-03 +	5.67E-03
50_8_4	5.58E-02 +	4.90E-02	1.27E-02*	1.14E-02	8.18E-03 +	7.28E-03
50_8_5	4.39E-02 +	3.86E-02	1.04E-02 +	1.15E-02	1.03E-02 +	9.14E-03
100_2_3	1.93E-02 +	1.70E-02	5.95E-03 +	6.60E-03	9.13E-03*	1.03E-02
100_2_4	3.22E-02 +	2.83E-02	8.46E-03 +	9.39E-03	8.08E-03 +	7.19E-03
100_2_5	4.52E-02 +	3.98E-02	1.06E-02 +	1.18E-02	1.71E-02 +	1.52E-02
100_5_3	3.62E-02 +	3.18E-02	5.96E-02 +	6.61E-02	1.12E-02 +	9.95E-03
100_5_4	1.70E-02 +	1.49E-02	1.44E-02 +	1.59E-02	3.74E-03 +	3.33E-03
100_5_5	3.81E-02 +	3.35E-02	1.27E-02 +	1.41E-02	6.10E-03 +	5.43E-03
100_8_3	2.23E-02 +	1.96E-02	4.76E-03 +	5.28E-03	6.08E-03 +	5.41E-03
100_8_4	3.49E-02 +	3.07E-02	1.24E-02 +	1.37E-02	8.01E-03 +	7.13E-03
100_8_5	5.75E-02 +	5.05E-02	1.30E-02 +	1.44E-02	7.04E-03 +	5.92E-03
150_2_3	1.41E-02 +	1.23E-02	3.44E-02 +	3.82E-02	8.91E-03 +	7.39E-03
150_2_4	1.35E-02 +	1.18E-02	5.77E-03 +	6.40E-03	5.78E-03 +	4.79E-03
150_2_5	2.89E-02 +	2.51E-02	4.50E-03*	4.06E-03	6.58E-03 +	5.46E-03
150_5_3	1.52E-02 +	1.32E-02	4.16E-02 +	4.61E-02	4.47E-03 +	3.70E-03
150_5_4	2.17E-02 +	1.89E-02	9.07E-03 +	1.01E-02	6.44E-03 +	5.34E-03
150_5_5	2.05E-02 +	1.78E-02	1.09E-02 +	1.21E-02	4.06E-03 +	3.37E-03
150_8_3	2.50E-02 +	2.18E-02	1.15E-02 +	1.28E-02	3.82E-03 +	3.17E-03
150_8_4	2.80E-02 +	2.43E-02	4.42E-03 +	4.90E-03	6.49E-03 +	5.38E-03
150_8_5	3.00E-02 +	2.61E-02	5.75E-03 +	6.38E-03	7.53E-03 +	6.25E-03
200_2_3	3.70E-02 +	3.26E-02	1.28E-02 +	1.42E-02	5.14E-03 +	4.27E-03
200_2_4	2.53E-02 +	2.22E-02	1.36E-02 +	1.50E-02	6.25E-03 +	5.19E-03
200_2_5	1.76E-02 +	1.55E-02	6.07E-02 +	6.73E-02	2.37E-02 +	1.97E-02
200_5_3	2.21E-02 +	1.95E-02	1.91E-02 +	2.12E-02	6.45E-03 +	5.35E-03
200_5_4	3.37E-02 +	2.97E-02	8.74E-03 +	9.69E-03	7.90E-03 +	6.55E-03
200_5_5	2.51E-02 +	2.21E-02	5.04E-02 +	5.59E-02	5.27E-03 +	4.37E-03
200_8_3	2.54E-02 +	2.24E-02	6.00E-03 +	6.65E-03	7.75E-03 +	6.43E-03
200_8_4	2.66E-02 +	2.34E-02	1.79E-02 +	1.98E-02	4.66E-03 +	3.87E-03
200_8_5	2.29E-02 +	2.01E-02	1.32E-02 +	1.46E-02	4.28E-03 +	3.55E-03
+/-/*	45/0/0		43/0/2		44/0/1	

knowledge-based inter-factory load balancing with Q-learning-based adaptive intra-factory VNS.

Comprehensive experiments on DHHFSP instances of varying scales show that HMOPSO-QLS significantly outperforms several classical multiobjective optimization algorithms in terms of convergence and solution diversity. In the context of distributed human-machine collaborative manufacturing, the proposed framework and algorithm support more effective configuration of factories, machines, and workers, producing robust schedules that are directly applicable to practical production decision-making and cross-factory coordination.

CRedit authorship contribution statement

Wenqiang Zhang: Writing – review & editing, Supervision, Project administration, Methodology, Funding acquisition, Formal analysis. **Sen Yang:** Writing – original draft, Visualization, Software, Methodology, Investigation. **Mingzhe Li:** Writing – original draft, Visualization, Validation, Software, Resources, Methodology, Investigation. **Yashuang Mu:** Validation, Methodology, Funding acquisition, Formal analysis. **Guohui Zhang:** Validation, Methodology, Funding acquisition, Conceptualization. **Mitsuo Gen:** Methodology, Formal analysis, Conceptualization. **Peng Li:** Supervision, Project administration, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This research work is supported by the National Natural Science Foundation of China (52575603, 62006071), the Key Research and Development Special Program of Henan Province, China (231111221200), the Natural Science Foundation of Henan Province, China (252300421884), and the Zhengzhou Science and Technology Collaborative Innovation Project, China (21ZZXTCX19).

Appendix. Detailed experimental results

This appendix provides the full detailed data for the experiments discussed in Section 5.

Table A.12

Index results of MOHPSO-LS1 and MOHPSO-LS2 for DHHFSP-WME.

Problems	IGD		HV		SP	
	MOHPSO-LS1	MOHPSO-LS2	MOHPSO-LS1	MOHPSO-LS2	MOHPSO-LS1	MOHPSO-LS2
20_2_3	1.14E-02+	1.04E-02	4.40E-02+	4.68E-02	1.10E-02+	1.01E-02
20_2_4	2.80E-02+	2.59E-02	1.28E-02+	1.40E-02	1.18E-02+	1.08E-02
20_2_5	5.61E-02*	5.67E-02	2.99E-02+	3.17E-02	7.37E-03+	6.93E-03
20_5_3	2.64E-02+	2.44E-02	6.41E-02+	6.89E-02	9.57E-03+	8.89E-03
20_5_4	3.06E-02+	2.83E-02	2.62E-03+	2.78E-03	9.06E-03+	8.33E-03
20_5_5	3.42E-02+	3.21E-02	1.26E-02+	1.34E-02	1.17E-02+	1.09E-02
20_8_3	1.71E-02+	1.62E-02	9.11E-03+	9.90E-03	8.77E-03+	8.12E-03
20_8_4	3.31E-02+	3.07E-02	7.31E-02+	7.77E-02	9.30E-03+	8.54E-03
20_8_5	5.35E-02+	5.02E-02	8.77E-03+	9.24E-03	2.03E-02+	1.88E-02
50_2_3	3.23E-02+	2.97E-02	5.70E-02+	6.12E-02	7.04E-03+	6.42E-03
50_2_4	1.37E-02+	1.26E-02	1.62E-02+	1.72E-02	1.65E-02+	1.54E-02
50_2_5	2.46E-02+	2.30E-02	5.32E-03+	5.72E-03	9.04E-03+	8.25E-03
50_5_3	2.40E-02+	2.25E-02	6.38E-02*	6.25E-02	6.67E-03+	6.17E-03
50_5_4	3.96E-02+	3.75E-02	7.69E-03+	8.34E-03	1.16E-02+	1.07E-02
50_5_5	3.83E-02*	3.84E-02	1.10E-02+	1.18E-02	7.40E-03+	6.74E-03
50_8_3	2.54E-02+	2.35E-02	9.52E-03+	1.03E-02	5.86E-03+	5.45E-03
50_8_4	5.16E-02+	4.76E-02	1.11E-02+	1.30E-02	7.64E-03+	6.99E-03
50_8_5	3.95E-02+	3.75E-02	1.11E-02*	1.01E-02	9.66E-03+	8.77E-03
100_2_3	1.74E-02+	1.65E-02	6.40E-03+	6.79E-03	1.07E-02+	8.76E-03
100_2_4	2.98E-02+	2.75E-02	8.89E-03+	9.66E-03	7.50E-03+	6.90E-03
100_2_5	4.74E-02*	4.76E-02	1.14E-02+	1.21E-02	1.61E-02+	1.46E-02
100_5_3	3.35E-02+	3.09E-02	6.25E-02+	6.80E-02	1.03E-02+	9.56E-03
100_5_4	1.55E-02+	1.45E-02	1.53E-02+	1.64E-02	3.42E-03+	3.20E-03
100_5_5	3.52E-02+	3.25E-02	1.37E-02+	1.45E-02	5.59E-03+	5.21E-03
100_8_3	2.06E-02+	1.90E-02	5.02E-03+	5.43E-03	5.52E-03+	5.19E-03
100_8_4	3.25E-02+	2.98E-02	1.33E-02+	1.41E-02	7.52E-03+	6.85E-03
100_8_5	5.29E-02+	4.91E-02	1.41E-02+	1.49E-02	6.07E-03+	5.68E-03
150_2_3	1.27E-02+	1.19E-02	3.62E-02+	3.91E-02	7.83E-03+	7.30E-03
150_2_4	1.23E-02+	1.14E-02	6.20E-03+	6.56E-03	4.96E-03+	4.73E-03
150_2_5	2.60E-02+	2.44E-02	3.97E-03+	4.61E-03	5.75E-03+	5.39E-03
150_5_3	1.39E-02+	1.29E-02	4.41E-02+	4.72E-02	4.92E-03*	4.96E-03
150_5_4	2.00E-02+	1.84E-02	9.65E-03+	1.03E-02	5.61E-03+	5.28E-03
150_5_5	1.86E-02+	1.73E-02	1.19E-02+	1.24E-02	3.49E-03+	3.33E-03
150_8_3	2.23E-02+	2.12E-02	1.25E-02+	1.31E-02	3.24E-03+	3.13E-03
150_8_4	2.50E-02+	2.37E-02	4.73E-03+	5.03E-03	5.68E-03+	5.32E-03
150_8_5	2.76E-02+	2.54E-02	6.20E-03+	6.54E-03	6.52E-03+	6.17E-03
200_2_3	3.35E-02+	3.13E-02	1.06E-02+	1.38E-02	4.52E-03+	4.22E-03
200_2_4	2.27E-02+	2.13E-02	1.47E-02+	1.54E-02	5.34E-03+	5.12E-03
200_2_5	1.62E-02+	1.49E-02	6.45E-02+	6.90E-02	2.03E-02+	1.94E-02
200_5_3	2.00E-02+	1.87E-02	2.02E-02+	2.18E-02	5.59E-03+	5.29E-03
200_5_4	3.12E-02+	2.85E-02	9.38E-03+	9.94E-03	6.81E-03+	6.47E-03
200_5_5	2.34E-02+	2.12E-02	5.41E-02+	5.73E-02	4.60E-03+	4.32E-03
200_8_3	2.29E-02+	2.15E-02	6.41E-03+	6.82E-03	6.63E-03+	6.35E-03
200_8_4	2.43E-02+	2.25E-02	1.92E-02+	2.03E-02	4.56E-03*	4.72E-03
200_8_5	2.11E-02+	1.93E-02	1.39E-02+	1.50E-02	3.74E-03+	3.51E-03
+/-/*	42/0/3		43/0/2		43/0/2	

Table A.13

Index results of MOHPSO-LS2 and HMOPSO-QLS for DHHFSP-WME.

Problems	IGD		HV		SP	
	MOHPSO-LS2	HMOPSO-QLS	MOHPSO-LS2	HMOPSO-QLS	MOHPSO-LS2	HMOPSO-QLS
20_2_3	1.10E-02+	1.00E-02	4.71E-02+	4.89E-02	1.04E-02+	9.67E-03
20_2_4	2.65E-02+	2.48E-02	1.38E-02+	1.46E-02	1.11E-02+	1.03E-02
20_2_5	5.93E-02+	4.67E-02	3.11E-02+	3.31E-02	7.33E-03+	6.64E-03
20_5_3	2.45E-02+	2.34E-02	6.83E-02+	7.19E-02	8.87E-03+	8.52E-03
20_5_4	2.93E-02+	2.71E-02	2.74E-03+	2.90E-03	8.54E-03+	7.98E-03
20_5_5	3.32E-02+	3.08E-02	1.31E-02+	1.40E-02	1.13E-02+	1.05E-02
20_8_3	1.67E-02+	1.55E-02	9.95E-03+	1.03E-02	8.10E-03+	7.78E-03
20_8_4	3.05E-02+	2.94E-02	7.58E-02+	8.11E-02	8.45E-03+	8.18E-03
20_8_5	5.01E-02+	4.81E-02	8.86E-03+	9.64E-03	1.90E-02+	1.80E-02
50_2_3	2.92E-02+	2.85E-02	5.86E-02+	6.39E-02	6.70E-03+	6.15E-03
50_2_4	1.27E-02+	1.21E-02	1.72E-02+	1.80E-02	1.56E-02+	1.47E-02
50_2_5	2.29E-02+	2.20E-02	5.68E-03+	5.96E-03	8.12E-03+	7.90E-03
50_5_3	2.23E-02+	2.16E-02	6.20E-02+	7.57E-02	6.29E-03+	5.92E-03
50_5_4	3.97E-02+	3.60E-02	7.88E-03+	8.70E-03	1.26E-02+	1.03E-02
50_5_5	3.90E-02+	2.72E-02	1.12E-02+	1.23E-02	7.08E-03+	6.46E-03

(continued on next page)

Table A.13 (continued).

Problems	IGD		HV		SP	
	MOHPSO-LS2	HMOPSO-QLS	MOHPSO-LS2	HMOPSO-QLS	MOHPSO-LS2	HMOPSO-QLS
50_8_3	2.36E-02+	2.25E-02	9.78E-03+	1.07E-02	5.41E-03+	5.22E-03
50_8_4	4.94E-02+	4.56E-02	1.26E-02+	1.36E-02	7.22E-03+	6.70E-03
50_8_5	3.68E-02+	3.59E-02	9.82E-03+	1.24E-02	8.69E-03+	8.41E-03
100_2_3	1.71E-02+	1.58E-02	6.57E-03+	7.09E-03	9.23E-03+	8.40E-03
100_2_4	2.89E-02+	2.63E-02	9.48E-03+	1.01E-02	6.94E-03+	6.61E-03
100_2_5	4.67E-02+	3.70E-02	1.20E-02+	1.26E-02	1.44E-02+	1.40E-02
100_5_3	3.10E-02+	2.96E-02	6.66E-02+	7.10E-02	9.66E-03+	9.16E-03
100_5_4	1.50E-02+	1.39E-02	1.65E-02+	1.71E-02	3.17E-03+	3.06E-03
100_5_5	3.27E-02+	3.12E-02	1.37E-02+	1.52E-02	5.45E-03+	5.00E-03
100_8_3	2.01E-02+	1.82E-02	5.30E-03+	5.67E-03	5.37E-03+	4.98E-03
100_8_4	2.95E-02+	2.86E-02	1.42E-02+	1.47E-02	7.17E-03+	6.56E-03
100_8_5	4.96E-02+	4.71E-02	1.49E-02+	1.55E-02	5.62E-03+	5.44E-03
150_2_3	1.21E-02+	1.14E-02	3.90E-02+	4.15E-02	7.53E-03+	6.89E-03
150_2_4	1.21E-02+	1.10E-02	6.48E-03+	6.95E-03	4.95E-03+	4.47E-03
150_2_5	2.42E-02+	2.34E-02	4.57E-03+	4.89E-03	5.51E-03+	5.08E-03
150_5_3	1.37E-02+	1.23E-02	4.69E-02+	5.01E-02	5.21E-03+	3.45E-03
150_5_4	1.94E-02+	1.76E-02	9.80E-03+	1.09E-02	5.27E-03+	4.97E-03
150_5_5	1.79E-02+	1.66E-02	1.20E-02+	1.32E-02	3.41E-03+	3.14E-03
150_8_3	2.18E-02+	2.03E-02	1.27E-02+	1.39E-02	3.23E-03+	2.96E-03
150_8_4	2.36E-02+	2.27E-02	4.78E-03+	5.33E-03	5.43E-03+	5.02E-03
150_8_5	2.49E-02+	2.43E-02	6.19E-03+	6.93E-03	6.11E-03+	5.82E-03
200_2_3	3.25E-02+	3.00E-02	1.07E-02+	1.54E-02	4.17E-03+	3.98E-03
200_2_4	2.19E-02+	2.05E-02	1.53E-02+	1.64E-02	5.27E-03+	4.83E-03
200_2_5	1.51E-02+	1.42E-02	6.68E-02+	7.32E-02	2.01E-02+	1.83E-02
200_5_3	1.97E-02+	1.79E-02	2.10E-02+	2.31E-02	5.27E-03+	4.99E-03
200_5_4	2.93E-02+	2.73E-02	9.58E-03+	1.05E-02	6.73E-03+	6.10E-03
200_5_5	2.16E-02+	2.03E-02	5.70E-02+	6.07E-02	4.36E-03+	4.07E-03
200_8_3	2.17E-02+	2.06E-02	6.80E-03+	7.23E-03	6.35E-03+	5.99E-03
200_8_4	2.28E-02+	2.16E-02	2.06E-02+	2.15E-02	4.81E-03+	3.60E-03
200_8_5	1.99E-02+	1.85E-02	1.46E-02+	1.59E-02	3.61E-03+	3.31E-03
+/-/*						
45/0/0						
45/0/0						
45/0/0						

Table A.14

IGD index results of all compared algorithms for DHHFSP-WME.

Problems	MOEA-LS	MOPSODS-DE	MOHEADE	SPEA2	NSGA-II	MOPSO	HMOPSO-QLS
20_2_3	1.67E-01 +	1.17E-01 +	2.05E-02 +	9.45E-02 +	8.88E-02 +	1.56E-01 +	9.81E-03
20_2_4	2.05E-01 +	7.47E-02 +	3.28E-02 +	8.81E-02 +	8.00E-02 +	1.02E-01 +	2.43E-02
20_2_5	2.64E-01 +	2.73E-01 +	1.02E-01 +	2.42E-01 +	2.26E-01 +	2.74E-01 +	4.58E-02
20_5_3	8.18E-02 +	6.77E-02 +	4.09E-02 +	6.85E-02 +	6.30E-02 +	9.81E-02 +	2.29E-02
20_5_4	1.20E-01 +	7.67E-02 +	4.50E-02 +	6.92E-02 +	6.49E-02 +	9.70E-02 +	2.65E-02
20_5_5	1.23E-01 +	1.80E-01 +	7.53E-02 +	1.91E-01 +	1.77E-01 +	2.12E-01 +	3.02E-02
20_8_3	7.60E-02 +	1.64E-01 +	4.28E-02 +	1.20E-01 +	1.18E-01 +	1.65E-01 +	1.52E-02
20_8_4	7.59E-02 +	6.33E-02 +	5.06E-02 +	6.90E-02 +	6.33E-02 +	8.93E-02 +	2.89E-02
20_8_5	8.82E-02 +	7.99E-02 +	6.29E-02 +	8.25E-02 +	7.68E-02 +	1.10E-01 +	4.72E-02
50_2_3	2.09E-01 +	2.46E-01 +	3.88E-02 +	1.38E-01 +	1.48E-01 +	2.04E-01 +	2.79E-02
50_2_4	3.67E-01 +	2.35E-01 +	4.00E-02 +	1.55E-01 +	1.60E-01 +	1.84E-01 +	1.18E-02
50_2_5	6.57E-02 +	4.24E-02 +	4.06E-02 +	5.83E-02 +	5.34E-02 +	7.76E-02 +	2.16E-02
50_5_3	1.09E-01 +	9.59E-02 +	5.00E-02 +	6.57E-02 +	7.11E-02 +	1.16E-01 +	2.11E-02
50_5_4	2.74E-01 +	1.28E-01 +	5.73E-02 +	9.90E-02 +	9.53E-02 +	1.43E-01 +	3.52E-02
50_5_5	9.45E-02 +	9.56E-02 +	6.09E-02 +	7.44E-02 +	7.27E-02 +	1.05E-01 +	2.67E-02
50_8_3	9.97E-02 +	8.01E-02 +	5.86E-02 +	5.58E-02 +	5.81E-02 +	9.71E-02 +	2.21E-02
50_8_4	1.00E-01 +	1.13E-01 +	8.13E-02 +	9.01E-02 +	7.96E-02 +	1.21E-01 +	4.47E-02
50_8_5	9.81E-02 +	1.12E-01 +	3.19E-02 *	8.92E-02 +	7.84E-02 +	1.25E-01 +	3.52E-02
100_2_3	1.19E-01 +	7.01E-02 +	2.88E-02 +	4.58E-02 +	4.99E-02 +	1.26E-01 +	1.55E-02
100_2_4	8.75E-02 +	1.35E-01 +	7.49E-02 +	9.77E-02 +	1.12E-01 +	1.63E-01 +	2.58E-02
100_2_5	2.44E-01 +	1.12E-01 +	1.07E-01 +	8.56E-02 +	1.10E-01 +	1.07E-01 +	3.63E-02
100_5_3	1.43E-01 +	8.79E-02 +	4.83E-02 +	4.95E-02 +	5.05E-02 +	1.18E-01 +	2.90E-02
100_5_4	9.35E-02 +	8.38E-02 +	6.31E-02 +	4.69E-02 +	5.91E-02 +	8.48E-02 +	1.36E-02
100_5_5	1.25E-01 +	1.46E-01 +	5.69E-02 +	1.14E-01 +	1.04E-01 +	1.80E-01 +	3.05E-02
100_8_3	6.64E-02 +	1.23E-01 +	3.11E-02 +	7.29E-02 +	6.66E-02 +	1.30E-01 +	1.79E-02
100_8_4	1.01E-01 +	1.10E-01 +	4.86E-02 +	7.89E-02 +	1.01E-01 +	1.32E-01 +	2.80E-02
100_8_5	8.38E-02 +	1.06E-01 +	3.17E-02 *	9.51E-02 +	8.66E-02 +	1.40E-01 +	4.61E-02
150_2_3	9.21E-02 +	6.60E-02 +	9.01E-02 +	3.02E-02 +	4.49E-02 +	7.01E-02 +	1.12E-02
150_2_4	1.01E-01 +	5.17E-02 +	2.90E-02 +	3.74E-02 +	4.59E-02 +	7.93E-02 +	1.07E-02

(continued on next page)

Table A.14 (continued).

Problems	MOEA-LS	MOPSODS-DE	MOHEADE	SPEA2	NSGA-II	MOPSO	HMOPSO-QLS
150_2_5	1.13E-01 +	1.09E-01 +	5.77E-02 +	7.97E-02 +	7.24E-02 +	1.14E-01 +	2.30E-02
150_5_3	1.39E-01 +	1.76E-01 +	4.04E-02 +	8.27E-02 +	9.98E-02 +	1.74E-01 +	1.21E-02
150_5_4	9.52E-02 +	1.77E-01 +	5.41E-02 +	7.64E-02 +	7.82E-02 +	1.47E-01 +	1.72E-02
150_5_5	8.73E-02 +	1.14E-01 +	4.22E-02 +	6.80E-02 +	6.54E-02 +	1.11E-01 +	1.63E-02
150_8_3	9.26E-02 +	1.16E-01 +	4.22E-02 +	6.15E-02 +	7.94E-02 +	1.21E-01 +	1.99E-02
150_8_4	1.03E-01 +	9.84E-02 +	5.19E-02 +	5.88E-02 +	7.23E-02 +	1.01E-01 +	2.22E-02
150_8_5	8.41E-02 +	1.40E-01 +	4.63E-02 +	1.07E-01 +	1.10E-01 +	1.65E-01 +	2.38E-02
200_2_3	1.40E-01 +	1.32E-01 +	4.76E-02 +	6.47E-02 +	7.53E-02 +	1.11E-01 +	2.94E-02
200_2_4	1.47E-01 +	7.27E-02 +	4.34E-02 +	4.50E-02 +	5.50E-02 +	8.16E-02 +	2.00E-02
200_2_5	8.85E-02 +	9.57E-02 +	4.54E-02 +	6.71E-02 +	5.84E-02 +	9.29E-02 +	1.40E-02
200_5_3	7.29E-02 +	1.14E-01 +	3.52E-02 +	5.65E-02 +	4.99E-02 +	1.32E-01 +	1.76E-02
200_5_4	9.42E-02 +	1.18E-01 +	4.25E-02 +	7.03E-02 +	8.08E-02 +	1.25E-01 +	2.68E-02
200_5_5	1.02E-01 +	1.39E-01 +	6.27E-02 +	6.93E-02 +	9.41E-02 +	1.36E-01 +	1.99E-02
200_8_3	6.63E-02 +	1.63E-01 +	3.36E-02 +	6.92E-02 +	9.11E-02 +	1.56E-01 +	2.02E-02
200_8_4	9.91E-02 +	1.29E-01 +	3.72E-02 +	7.22E-02 +	8.75E-02 +	1.44E-01 +	2.11E-02
200_8_5	9.12E-02 +	1.01E-01 +	4.07E-02 +	7.16E-02 +	6.41E-02 +	1.18E-01 +	1.81E-02
+/-/*	45/0/0	44/0/1	43/0/2	45/0/0	45/0/0	45/0/0	

Table A.15*HV* index results of all compared algorithms for DHHFSP-WME.

Problems	MOEA-LS	MOPSODS-DE	MOHEADE	SPEA2	NSGA-II	MOPSO	HMOPSO-QLS
20_2_3	1.35E-02 +	1.61E-02 +	4.41E-02 *	1.92E-02 +	2.29E-02 +	1.37E-02 +	4.93E-02
20_2_4	7.88E-03 +	6.70E-03 +	1.19E-02 +	6.95E-03 +	6.94E-03 +	7.03E-03 +	1.47E-02
20_2_5	1.13E-02 +	4.67E-03 +	2.11E-02 +	5.35E-03 +	6.60E-03 +	6.96E-03 +	3.34E-02
20_5_3	1.14E-02 +	7.45E-03 +	5.75E-03 +	2.73E-03 +	2.52E-03 +	5.68E-03 +	7.26E-02
20_5_4	1.13E-02 *	4.60E-03 +	7.26E-03 +	1.37E-03 +	2.78E-03 +	4.50E-03 +	2.93E-03
20_5_5	5.05E-03 +	2.37E-03 +	9.10E-03 +	9.90E-04 +	1.88E-03 +	1.45E-03 +	1.41E-02
20_8_3	9.92E-03 +	4.33E-03 +	7.35E-03 +	2.63E-03 +	3.50E-03 +	3.65E-03 +	1.04E-02
20_8_4	3.73E-03 +	3.17E-03 +	2.84E-03 +	1.10E-03 +	1.98E-04 +	1.97E-03 +	8.19E-02
20_8_5	8.36E-03 +	6.92E-03 +	7.19E-03 +	3.55E-03 +	2.50E-03 +	5.40E-03 +	9.74E-03
50_2_3	4.06E-02 +	2.53E-02 +	5.28E-02 +	3.09E-02 +	3.19E-02 +	2.96E-02 +	6.45E-02
50_2_4	4.90E-03 +	3.06E-03 +	1.51E-02 +	5.56E-03 +	5.60E-03 +	4.63E-03 +	1.81E-02
50_2_5	4.58E-03 +	5.54E-03 *	5.25E-03 +	1.39E-03 +	1.97E-03 +	4.30E-03 +	5.96E-03
50_5_3	1.90E-02 +	1.44E-02 +	1.54E-02 +	3.52E-03 +	3.25E-03 +	1.24E-02 +	7.64E-02
50_5_4	3.67E-03 +	4.25E-03 +	6.24E-03 +	3.11E-03 +	2.97E-03 +	4.35E-03 +	8.70E-03
50_5_5	7.16E-03 +	5.70E-03 +	5.93E-03 +	3.67E-04 +	7.69E-04 +	3.70E-03 +	1.24E-02
50_8_3	5.68E-03 +	3.98E-03 +	1.00E-02 +	2.38E-03 +	4.21E-03 +	3.46E-03 +	1.07E-02
50_8_4	1.17E-02 +	1.10E-02 +	1.29E-02 *	1.56E-03 +	2.02E-03 +	8.12E-03 +	1.38E-02
50_8_5	8.94E-03 +	7.26E-03 +	1.05E-02 +	4.53E-03 +	3.38E-03 +	6.94E-03 +	1.25E-02
100_2_3	5.56E-03 +	2.67E-03 +	4.99E-03 +	4.16E-03 +	4.36E-03 +	2.89E-03 +	7.08E-03
100_2_4	5.24E-03 +	5.28E-03 +	6.96E-03 +	6.61E-03 +	6.50E-03 +	4.12E-03 +	1.02E-02
100_2_5	1.81E-02 *	8.84E-03 +	1.03E-02 +	5.64E-03 +	6.17E-03 +	6.74E-03 +	1.26E-02
100_5_3	2.07E-03 +	1.37E-03 +	1.30E-03 +	2.05E-04 +	1.90E-04 +	6.67E-04 +	7.17E-02
100_5_4	3.46E-03 +	8.32E-04 +	6.57E-03 +	3.23E-03 +	2.74E-04 +	2.76E-03 +	1.71E-02
100_5_5	1.45E-02 +	9.39E-03 +	1.28E-02 +	6.91E-03 +	7.05E-03 +	6.47E-03 +	1.53E-02
100_8_3	2.55E-03 +	3.59E-03 +	5.04E-03 +	1.38E-03 +	1.71E-03 +	2.99E-03 +	5.72E-03
100_8_4	1.44E-03 +	7.09E-03 +	1.06E-02 +	1.42E-03 +	3.65E-03 +	6.41E-03 +	1.47E-02
100_8_5	6.46E-03 +	1.00E-02 +	1.24E-02 +	3.31E-03 +	4.33E-03 +	7.62E-03 +	1.57E-02
150_2_3	3.51E-03 +	2.47E-03 +	9.68E-03 +	5.75E-03 +	2.40E-03 +	4.15E-03 +	4.19E-02
150_2_4	1.93E-03 +	1.65E-03 +	6.78E-03 +	3.55E-03 +	2.52E-03 +	8.66E-04 +	6.95E-03
150_2_5	3.25E-03 +	2.92E-03 +	2.92E-03 +	1.28E-03 +	1.54E-03 +	2.63E-03 +	4.94E-03
150_5_3	6.47E-03 +	3.84E-03 +	8.16E-03 +	4.94E-03 +	6.88E-03 +	5.20E-03 +	5.06E-02
150_5_4	1.27E-02 +	6.55E-03 +	1.29E-02 *	8.19E-03 +	8.77E-03 +	8.14E-03 +	1.10E-02
150_5_5	1.13E-03 +	7.58E-03 +	9.64E-03 +	4.29E-03 +	3.62E-03 +	7.55E-03 +	1.33E-02
150_8_3	2.78E-03 +	5.47E-03 +	9.36E-03 +	4.04E-03 +	6.66E-03 +	7.59E-03 +	1.40E-02
150_8_4	1.87E-03 +	2.95E-03 +	3.56E-03 +	5.33E-04 +	1.55E-03 +	2.31E-03 +	5.33E-03
150_8_5	6.43E-03 +	3.44E-03 +	4.22E-03 +	2.36E-03 +	3.51E-03 +	3.19E-03 +	7.00E-03
200_2_3	1.34E-02 +	9.86E-03 +	1.17E-02 +	1.06E-02 +	1.36E-02 +	1.26E-02 +	1.56E-02
200_2_4	5.37E-03 +	9.33E-03 +	1.12E-02 +	1.07E-02 +	9.41E-03 +	9.12E-03 +	1.65E-02
200_2_5	1.62E-02 +	1.17E-02 +	1.23E-02 +	6.24E-03 +	8.38E-03 +	1.55E-02 +	7.31E-02
200_5_3	1.93E-02 +	1.37E-02 +	1.63E-02 +	1.64E-02 +	1.67E-02 +	1.53E-02 +	2.33E-02
200_5_4	8.19E-03 +	8.24E-03 +	1.01E-02 *	7.62E-03 +	7.23E-03 +	7.77E-03 +	1.06E-02
200_5_5	7.48E-03 +	3.91E-03 +	6.16E-03 +	1.81E-03 +	3.18E-03 +	4.25E-03 +	6.13E-02
200_8_3	1.52E-02 +	6.26E-03 +	1.54E-02 *	5.83E-03 +	9.05E-03 +	8.42E-03 +	1.12E-02
200_8_4	1.39E-02 +	9.54E-03 +	1.74E-02 +	8.51E-03 +	1.02E-02 +	9.95E-03 +	2.18E-02
200_8_5	9.39E-03 +	9.77E-03 +	1.03E-02 +	6.80E-03 +	5.92E-03 +	9.39E-03 +	1.59E-02
+/-/*	43/0/2	44/0/1	40/0/5	45/0/0	45/0/0	45/0/0	

Table A.16*SP* index results of all compared algorithms for DHHFSP-WME.

Problems	MOEA-LS	MOPSODS-DE	MOHEADE	SPEA2	NSGA-II	MOPSO	HMOPSO-QLS
20_2_3	1.09E-02 +	2.37E-02 +	1.23E-02 +	2.97E-02 +	2.38E-02 +	1.59E-02 +	9.57E-03
20_2_4	1.45E-02 +	1.38E-02 +	1.80E-02 +	2.59E-02 +	2.38E-02 +	2.91E-02 +	1.02E-02
20_2_5	1.14E-02 +	1.07E-02 +	3.48E-02 +	3.34E-02 +	4.66E-02 +	1.93E-02 +	6.58E-03
20_5_3	2.90E-02 +	1.20E-02 +	1.98E-02 +	3.22E-02 +	3.37E-02 +	2.55E-02 +	8.44E-03
20_5_4	2.61E-02 +	1.16E-02 +	2.01E-02 +	2.20E-02 +	2.42E-02 +	1.44E-02 +	7.90E-03
20_5_5	2.77E-02 +	1.23E-02 *	2.08E-02 +	2.49E-02 +	2.38E-02 +	2.27E-02 +	1.04E-02
20_8_3	3.34E-02 +	8.45E-03 +	1.78E-02 +	3.00E-02 +	2.69E-02 +	2.52E-02 +	7.70E-03
20_8_4	2.74E-02 +	1.25E-02 +	1.92E-02 +	2.45E-02 +	2.78E-02 +	3.20E-02 +	8.10E-03
20_8_5	1.69E-02 +	1.20E-02 *	1.56E-02 +	2.75E-02 +	2.62E-02 +	1.74E-02 +	1.78E-02
50_2_3	4.12E-02 +	3.59E-02 +	5.75E-02 +	6.05E-02 +	4.46E-02 +	3.69E-02 +	6.09E-03
50_2_4	1.32E-02 *	1.85E-02 +	2.32E-02 +	3.16E-02 +	3.39E-02 +	4.10E-02 +	1.46E-02
50_2_5	2.58E-02 +	1.34E-02 +	2.37E-02 +	3.00E-02 +	3.66E-02 +	1.52E-02 +	7.82E-03
50_5_3	5.47E-02 +	1.61E-02 -	2.30E-02 +	4.25E-02 +	4.19E-02 +	1.93E-02 +	3.86E-02
50_5_4	3.79E-02 +	2.71E-02 +	2.53E-02 +	3.38E-02 +	4.19E-02 +	3.41E-02 +	1.01E-02
50_5_5	3.45E-02 +	1.26E-02 +	1.90E-02 +	3.27E-02 +	3.33E-02 +	1.65E-02 +	6.39E-03
50_8_3	3.44E-02 +	1.91E-02 +	1.75E-02 +	3.50E-02 +	4.13E-02 +	1.91E-02 +	5.17E-03
50_8_4	1.19E-02 +	1.44E-02 +	2.28E-02 +	3.16E-02 +	4.30E-02 +	1.69E-02 +	6.63E-03
50_8_5	1.43E-02 +	1.14E-02 +	2.07E-02 +	3.19E-02 +	3.79E-02 +	2.22E-02 +	8.32E-03
100_2_3	4.22E-02 +	2.23E-02 +	3.89E-02 +	4.07E-02 +	3.45E-02 +	1.04E-02 +	8.48E-03
100_2_4	6.42E-02 +	1.88E-02 +	7.19E-02 +	4.89E-02 +	3.62E-02 +	2.29E-02 +	6.68E-03
100_2_5	3.63E-02 +	2.69E-02 +	3.53E-02 +	4.83E-02 +	3.95E-02 +	3.64E-02 +	1.41E-02
100_5_3	4.16E-02 +	1.95E-02 +	2.53E-02 +	3.58E-02 +	3.50E-02 +	1.41E-02 +	9.25E-03
100_5_4	4.09E-02 +	2.56E-02 +	2.34E-02 +	4.83E-02 +	3.73E-02 +	2.21E-02 +	3.10E-03
100_5_5	3.91E-02 +	2.44E-02 +	3.22E-02 +	4.07E-02 +	3.96E-02 +	2.76E-02 +	5.05E-03
100_8_3	3.68E-02 +	1.68E-02 +	2.17E-02 +	3.88E-02 +	4.15E-02 +	2.08E-02 +	5.03E-03
100_8_4	3.38E-02 +	1.97E-02 +	2.41E-02 +	3.86E-02 +	2.98E-02 +	2.64E-02 +	6.63E-03
100_8_5	2.25E-02 +	1.39E-02 +	2.07E-02 +	3.57E-02 +	3.48E-02 +	1.87E-02 +	5.50E-03
150_2_3	5.11E-02 +	2.57E-02 +	4.71E-02 +	6.42E-02 +	5.09E-02 +	3.17E-02 +	6.95E-03
150_2_4	3.63E-02 +	2.19E-02 +	3.58E-02 +	4.58E-02 +	4.23E-02 +	1.93E-02 +	4.51E-03
150_2_5	4.46E-02 +	2.29E-02 +	4.11E-02 +	4.25E-02 +	4.63E-02 +	2.53E-02 +	5.14E-03
150_5_3	4.75E-02 +	2.49E-02 +	5.93E-02 +	5.65E-02 +	4.10E-02 +	3.36E-02 +	3.49E-03
150_5_4	4.12E-02 +	2.39E-02 +	2.18E-02 +	3.71E-02 +	3.52E-02 +	3.06E-02 +	5.02E-03
150_5_5	4.77E-02 +	1.83E-02 +	2.36E-02 +	4.81E-02 +	4.53E-02 +	1.74E-02 +	3.17E-03
150_8_3	6.30E-02 +	2.94E-02 +	2.02E-02 +	4.87E-02 +	4.11E-02 +	1.86E-02 +	2.98E-03
150_8_4	3.87E-02 +	1.28E-02 +	1.67E-02 +	3.63E-02 +	2.89E-02 +	1.90E-02 +	5.07E-03
150_8_5	3.39E-02 +	1.54E-02 +	2.28E-02 +	3.07E-02 +	2.59E-02 +	1.92E-02 +	5.88E-03
200_2_3	6.40E-02 +	2.67E-02 +	5.36E-02 +	5.78E-02 +	4.82E-02 +	3.16E-02 +	4.02E-03
200_2_4	3.87E-02 +	1.65E-02 +	4.52E-02 +	2.86E-02 +	1.62E-02 +	2.62E-02 +	4.88E-03
200_2_5	4.54E-02 +	3.71E-02 +	3.86E-02 +	5.38E-02 +	6.03E-02 +	3.07E-02 +	1.85E-02
200_5_3	3.36E-02 +	2.79E-02 +	2.28E-02 +	3.80E-02 +	4.04E-02 +	2.27E-02 +	5.04E-03
200_5_4	4.95E-02 +	1.92E-02 +	3.41E-02 +	4.69E-02 +	4.41E-02 +	1.98E-02 +	6.17E-03
200_5_5	4.05E-02 +	1.91E-02 +	2.52E-02 +	4.41E-02 +	3.72E-02 +	2.02E-02 +	4.11E-03
200_8_3	4.75E-02 +	2.91E-02 +	2.29E-02 +	4.85E-02 +	4.01E-02 +	3.23E-02 +	6.05E-03
200_8_4	3.25E-02 +	2.69E-02 +	1.98E-02 +	3.81E-02 +	3.81E-02 +	2.92E-02 +	3.64E-03
200_8_5	3.53E-02 +	2.42E-02 +	1.45E-02 +	3.99E-02 +	4.90E-02 +	1.98E-02 +	3.34E-03
+/-/*	44/0/1	42/1/2	45/0/0	45/0/0	45/0/0	45/0/0	

Table A.17*C* index results of all compared algorithms for DHHFSP-WME.

A: HMOPSO-QLS B: MOEA-LS D: MOHEADE E: MOPSODS-DE F: NSGA-II G: SPEA2 H: MOPSO												
Problems	C(A,B)	C(B,A)	C(A,D)	C(D,A)	C(A,E)	C(E,A)	C(A,F)	C(F,A)	C(A,G)	C(G,A)	C(A,H)	C(H,A)
20_2_3	1.00	0.00	1.00	0.00	1.00	0.00	1.00	0.00	1.00	0.00	1.00	0.00
20_2_4	0.97	0.00	0.88	0.12	1.00	0.00	1.00	0.00	1.00	0.00	0.99	0.00
20_2_5	1.00	0.00	1.00	0.00	1.00	0.00	1.00	0.00	1.00	0.00	1.00	0.00
20_5_3	0.96	0.03	0.88	0.13	0.98	0.11	0.99	0.00	1.00	0.00	1.00	0.00
20_5_4	0.97	0.00	0.86	0.03	0.97	0.12	0.91	0.02	0.99	0.01	0.98	0.00
20_5_5	1.00	0.00	1.00	0.00	1.00	0.00	1.00	0.00	1.00	0.00	1.00	0.00
20_8_3	0.98	0.00	1.00	0.00	1.00	0.00	1.00	0.00	1.00	0.00	0.99	0.00
20_8_4	0.94	0.03	0.81	0.11	0.99	0.02	1.00	0.00	0.98	0.00	1.00	0.00
20_8_5	1.00	0.00	0.84	0.14	1.00	0.00	0.98	0.00	1.00	0.00	1.00	0.00
50_2_3	1.00	0.00	0.85	0.00	1.00	0.00	1.00	0.00	1.00	0.00	0.99	0.00
50_2_4	1.00	0.00	0.92	0.00	1.00	0.00	1.00	0.00	1.00	0.00	1.00	0.00
50_2_5	0.94	0.07	0.98	0.00	1.00	0.00	0.95	0.00	1.00	0.00	1.00	0.00
50_5_3	0.97	0.00	0.90	0.02	1.00	0.00	0.97	0.01	0.97	0.02	1.00	0.00

(continued on next page)

Table A.17 (continued).

A: HMOPSO-QLS B: MOEA-LS D: MOHEADE E: MOPSODS-DE F: NSGA-II G: SPEA2 H: MOPSO												
Problems	C(A,B)	C(B,A)	C(A,D)	C(D,A)	C(A,E)	C(E,A)	C(A,F)	C(F,A)	C(A,G)	C(G,A)	C(A,H)	C(H,A)
50_5_4	0.99	0.00	0.99	0.00	1.00	0.00	1.00	0.00	1.00	0.00	1.00	0.00
50_5_5	0.82	0.10	1.00	0.00	1.00	0.00	1.00	0.00	1.00	0.00	1.00	0.00
50_8_3	1.00	0.00	0.72	0.00	0.95	0.03	1.00	0.00	0.97	0.00	0.94	0.01
50_8_4	1.00	0.00	1.00	0.00	1.00	0.00	0.99	0.00	1.00	0.00	1.00	0.00
50_8_5	0.98	0.01	0.93	0.01	1.00	0.00	1.00	0.00	1.00	0.00	0.93	0.00
100_2_3	1.00	0.00	0.88	0.02	1.00	0.00	1.00	0.00	1.00	0.00	1.00	0.00
100_2_4	1.00	0.00	1.00	0.00	1.00	0.00	1.00	0.00	1.00	0.00	1.00	0.00
100_2_5	0.95	0.00	0.88	0.08	1.00	0.00	1.00	0.00	1.00	0.00	1.00	0.00
100_5_3	1.00	0.00	0.92	0.10	0.97	0.00	1.00	0.00	0.99	0.00	0.92	0.01
100_5_4	1.00	0.00	0.89	0.09	0.87	0.01	1.00	0.00	1.00	0.00	1.00	0.00
100_5_5	1.00	0.00	1.00	0.00	1.00	0.00	1.00	0.00	1.00	0.00	1.00	0.00
100_8_3	1.00	0.00	0.97	0.05	0.92	0.01	1.00	0.00	1.00	0.00	0.94	0.00
100_8_4	1.00	0.00	0.78	0.05	0.95	0.00	1.00	0.00	1.00	0.00	1.00	0.00
100_8_5	1.00	0.00	0.88	0.05	0.92	0.06	0.92	0.01	1.00	0.00	1.00	0.00
150_2_3	0.95	0.06	0.99	0.02	0.87	0.04	0.93	0.01	0.91	0.00	1.00	0.00
150_2_4	0.94	0.00	0.90	0.01	0.94	0.06	0.98	0.01	0.99	0.00	0.91	0.01
150_2_5	1.00	0.00	1.00	0.00	1.00	0.00	1.00	0.00	1.00	0.00	1.00	0.00
150_5_3	0.98	0.00	0.65	0.18	1.00	0.00	1.00	0.00	1.00	0.00	1.00	0.00
150_5_4	0.86	0.01	0.96	0.05	1.00	0.00	1.00	0.00	1.00	0.00	1.00	0.00
150_5_5	1.00	0.00	1.00	0.00	0.99	0.00	1.00	0.00	1.00	0.00	1.00	0.00
150_8_3	0.98	0.07	1.00	0.00	1.00	0.00	1.00	0.00	1.00	0.00	1.00	0.00
150_8_4	1.00	0.00	0.89	0.00	0.95	0.01	1.00	0.00	0.99	0.00	1.00	0.00
150_8_5	0.93	0.00	1.00	0.00	1.00	0.00	1.00	0.00	1.00	0.00	1.00	0.00
200_2_3	0.94	0.00	0.72	0.14	1.00	0.00	1.00	0.00	1.00	0.00	1.00	0.00
200_2_4	1.00	0.00	0.76	0.06	1.00	0.00	1.00	0.00	1.00	0.00	1.00	0.00
200_2_5	0.97	0.00	1.00	0.00	1.00	0.00	1.00	0.00	1.00	0.00	0.82	0.06
200_5_3	1.00	0.00	0.86	0.05	0.94	0.00	1.00	0.00	1.00	0.00	1.00	0.00
200_5_4	0.99	0.00	0.92	0.00	1.00	0.00	1.00	0.00	1.00	0.00	1.00	0.00
200_5_5	0.97	0.03	0.81	0.08	0.98	0.02	1.00	0.00	1.00	0.00	1.00	0.00
200_8_3	0.99	0.01	1.00	0.00	1.00	0.00	1.00	0.00	1.00	0.00	0.92	0.00
200_8_4	1.00	0.00	1.00	0.00	0.97	0.00	1.00	0.00	1.00	0.00	1.00	0.00
200_8_5	0.94	0.00	0.98	0.04	0.95	0.00	1.00	0.00	1.00	0.00	1.00	0.00

References

- [1] Duan J, Liu F, Zhang Q, Qin J, Zhou Y. Genetic programming hyper-heuristic-based solution for dynamic energy-efficient scheduling of hybrid flow shop scheduling with machine breakdowns and random job arrivals. *Expert Syst Appl* 2024;124375.
- [2] Yu F, Lu C, Zhou J, Yin L. Mathematical model and knowledge-based iterated greedy algorithm for distributed assembly hybrid flow shop scheduling problem with dual-resource constraints. *Expert Syst Appl* 2024;239:122434.
- [3] Leng J, Zhu X, Huang Z, Li X, Zheng P, Zhou X, Mourtzis D, Wang B, Qi Q, Shao H, et al. Unlocking the power of industrial artificial intelligence towards industry 5.0: Insights, pathways, and challenges. *J Manuf Syst* 2024;73:349–63.
- [4] Papacharalampopoulos A, Toupalis M, Lagios P, Karagianni OM, Stavropoulos P. Acquiring feedback into the manufacturing process design from industrial policies. *Procedia CIRP* 2024;128:579–84.
- [5] Gao Q, Jianhua L, Liu Shimin aZC. From human-related to human-centric: A review of shop floor scheduling problem under industry 5.0. *J Manuf Syst* 2025;82:531–46.
- [6] Li M, Wang G-G. A review of green shop scheduling problem. *Inform Sci* 2022;589:478–96.
- [7] Fatima T, Li B, Malik SA, Zhang D. The spatial effect of industrial intelligence on high-quality green development of industry under environmental regulations and low carbon intensity. *Sustain*. 2023;15(3):1903.
- [8] Gao L, Pan Q-K. A shuffled multi-swarm micro-migrating birds optimizer for a multi-resource-constrained flexible job shop scheduling problem. *Inform Sci* 2016;372:655–76.
- [9] Song H, Li J, Du Z, Yu X, Xu Y, Zheng Z, Li J. A Q-learning driven multi-objective evolutionary algorithm for worker fatigue dual-resource-constrained distributed hybrid flow shop. *Comput Oper Res* 2025;175:106919.
- [10] Zhang Q, Shao W, Shao Z, Pi D, Gao J. Deep reinforcement learning driven trajectory-based meta-heuristic for distributed heterogeneous flexible job shop scheduling problem. *Swarm Evol Comput* 2024;91:101753.
- [11] Lu C, Gao L, Yi J, Li X. Energy-efficient scheduling of distributed flow shop with heterogeneous factories: A real-world case from automobile industry in China. *IEEE Trans Ind Inform*. 2020;17(10):6687–96.
- [12] Zheng J, Wang L, Wang J-j. A cooperative coevolution algorithm for multi-objective fuzzy distributed hybrid flow shop. *Knowl-Based Syst* 2020;194:105536.
- [13] Cai J, Lei D, Li M. A shuffled frog-leaping algorithm with memplex quality for bi-objective distributed scheduling in hybrid flow shop. *Int J Prod Res* 2021;59(18):5404–21.
- [14] Jiang E, Wang L, Wang J. Decomposition-based multi-objective optimization for energy-aware distributed hybrid flow shop scheduling with multiprocessor tasks. *Tsinghua Sci Technol* 2021;26(5):646–63.
- [15] Shao W, Shao Z, Pi D. An ant colony optimization behavior-based MOEA/D for distributed heterogeneous hybrid flow shop scheduling problem under nonidentical time-of-use electricity tariffs. *IEEE Trans Autom Sci Eng* 2021;19(4):3379–94.
- [16] Wang J-J, Wang L. A cooperative memetic algorithm with learning-based agent for energy-aware distributed hybrid flow-shop scheduling. *IEEE Trans Evol Comput* 2021;26(3):461–75.
- [17] Lu C, Liu Q, Zhang B, Yin L. A Pareto-based hybrid iterated greedy algorithm for energy-efficient scheduling of distributed hybrid flowshop. *Expert Syst Appl* 2022;204:117555.
- [18] Cui H, Li X, Gao L, Zhang C. Multi-population genetic algorithm with greedy job insertion inter-factory neighbourhoods for multi-objective distributed hybrid flow-shop scheduling with unrelated-parallel machines considering tardiness. *Int J Prod Res* 2024;62(12):4427–45.
- [19] Sun X, Shen W, Fan J, Vogel-Heuser B, Zhang C. An improved non-dominated sorting genetic algorithm II for distributed heterogeneous hybrid flow-shop scheduling with blocking constraints. *J Manuf Syst* 2024;77:990–1008.
- [20] Cai J, Wang L. A shuffled frog leaping algorithm with q-learning for distributed hybrid flow shop scheduling problem with energy-saving. *J Artif Intell Soft Comput Res* 2024;14.
- [21] Zhang W, Li C, Gen M, Yang W, Zhang G. A multiobjective memetic algorithm with particle swarm optimization and Q-learning-based local search for energy-efficient distributed heterogeneous hybrid flow-shop scheduling problem. *Expert Syst Appl* 2024;237:121570.
- [22] Lu C, Zhou J, Gao L, Li X, Wang J. Modeling and multi-objective optimization for energy-aware scheduling of distributed hybrid flow-shop. *Appl Soft Comput* 2024;156:111508.
- [23] Zhao F, Yin F, Wang L, Yu Y. A co-evolution algorithm with dueling reinforcement learning mechanism for the energy-aware distributed heterogeneous flexible flow-shop scheduling problem. *IEEE Trans Syst Man, Cybern: Syst* 2024.
- [24] Zhao F, Yin F, Zhang J, Xu TP. A learning-based co-evolution optimization framework for energy-aware distributed heterogeneous flexible flow shop lot-streaming scheduling problem. *Expert Syst Appl* 2026;296:128986.
- [25] Lei D, Su B. A multi-class teaching-learning-based optimization for multi-objective distributed hybrid flow shop scheduling. *Knowl-Based Syst* 2023;263:110252.
- [26] Chen X, Li Y, Wang K, Wang L, Liu J, Wang J, Wang XV. Reinforcement learning for distributed hybrid flowshop scheduling problem with variable task splitting towards mass personalized manufacturing. *J Manuf Syst* 2024;76:188–206.

- [27] Cai J, Lei D. A cooperated shuffled frog-leaping algorithm for distributed energy-efficient hybrid flow shop scheduling with fuzzy processing time. *Complex Intell Syst* 2021;7(5):2235–53.
- [28] Deng L, Di Y, Wang L. A reinforcement-learning-based 3-D estimation of distribution algorithm for fuzzy distributed hybrid flow-shop scheduling considering on-time-delivery. *IEEE Trans Cybern* 2023;54(2):1024–36.
- [29] Shao W, Shao Z, Pi D. Multi-objective evolutionary algorithm based on multiple neighborhoods local search for multi-objective distributed hybrid flow shop scheduling problem. *Expert Syst Appl* 2021;183:115453.
- [30] Shao W, Shao Z, Pi D. A multi-neighborhood-based multi-objective memetic algorithm for the energy-efficient distributed flexible flow shop scheduling problem. *Neural Comput Appl* 2022;34(24):22303–30.
- [31] Wu X, Xie Z. Improved MOEA/D with local search for solving multi-stage distributed reentrant hybrid flow shop scheduling problem. *Expert Syst Appl* 2023;232:120893.
- [32] Sun X, Shen W, Fan J, Vogel-Heuser B, Bi F, Zhang C. Deep reinforcement learning-based multi-objective scheduling for distributed heterogeneous hybrid flow shops with blocking constraints. *Eng.* 2025;46:278–91.
- [33] Xuan H, Zheng Q-Q, Lv L, Li B. Q-learning based estimation of distribution algorithm for scheduling distributed heterogeneous flexible flow-shop with mixed buffering limitation. *Eng Appl Artif Intell* 2025;149:110537.
- [34] Shao W, Shao Z, Pi D. A network memetic algorithm for energy and labor-aware distributed heterogeneous hybrid flow shop scheduling problem. *Swarm Evol Comput* 2022;75:101190.
- [35] Gao Q, Hu H, Liu J, Zhuang C, Ding H, Li H. Distributed assembly shop scheduling problem for complex products considering multiskilled worker assignment and transportation time. *Int J Prod Res* 2024;1–23.
- [36] Deng L, Qiu Y, Gong W, Di Y, Li C. A dynamic decision-driven memetic algorithm for fuzzy distributed hybrid flow shop rescheduling considering quality control. *Expert Syst Appl* 2024;257:125002.
- [37] Xu T, Zhao F, Zhang J, Tang J, Zhou H. A feature based neural network model for distributed flexible flow shop scheduling considering worker and transportation factors. *Comput Ind Eng* 2025;110917.
- [38] Yu F, Lu C, Yin L, Zhou J. Modeling and optimization algorithm for energy-efficient distributed assembly hybrid flowshop scheduling problem considering worker resources. *J Ind Inf Integr* 2024;40:100620.
- [39] Gong G, Deng Q, Gong X, Huang D. A non-dominated ensemble fitness ranking algorithm for multi-objective flexible job-shop scheduling problem considering worker flexibility and green factors. *Knowl-Based Syst* 2021;231:107430.
- [40] Fernandez-Viagas V, Perez-Gonzalez P, Framinan JM. Efficiency of the solution representations for the hybrid flow shop scheduling problem with makespan objective. *Comput Oper Res* 2019;109:77–88.
- [41] Ruiz R, Maroto C. A genetic algorithm for hybrid flowshops with sequence dependent setup times and machine eligibility. *European J Oper Res* 2006;169(3):781–800.
- [42] Brah SA, Loo LL. Heuristics for scheduling in a flow shop with multiple processors. *European J Oper Res* 1999;113(1):113–22.
- [43] Zhang W, Gen M, Jo J. Hybrid sampling strategy-based multiobjective evolutionary algorithm for process planning and scheduling problem. *J Intell Manuf* 2014;25(5):881–97.
- [44] Zhang W, Li C, Gen M, Yang W, Zhang Z, Zhang G. Multiobjective particle swarm optimization with direction search and differential evolution for distributed flow-shop scheduling problem. *Math Biosci Eng* 2022;19(9):8833–65.
- [45] Zhang W, Wang Y, Yang Y, Gen M. Hybrid multiobjective evolutionary algorithm based on differential evolution for flow shop scheduling problems. *Comput Ind Eng* 2019;130:661–70.
- [46] Coello CAC, Pulido GT, Lechuga MS. Handling multiple objectives with particle swarm optimization. *IEEE Trans Evol Comput* 2004;8(3):256–79.
- [47] Kalyanmoy D. A fast and elitist multi-objective genetic algorithm: NSGA-II. *IEEE Trans Evol Comput* 2002;6(2):182–97.
- [48] Zitzler E, Laumanns M, Thiele L. SPEA2: Improving the strength Pareto evolutionary algorithm. *TIK Rep* 2001;103.